Endpoint-Regular Fixed-Point Universality

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Mechanical extraction — proof audit pending

Abstract

For uniformly endpoint-regularly varying terminal rate arrays, this paper proves a logarithmic Poisson point process for fixed points, identifies the fixed-point factorial moments and mean constant, and derives endpoint displacement, mesoscopic tangent-measure, multiplicative-kernel, and microscopic endpoint-permutation limits. The proof develops endpoint-specific saddle geometry, anisotropic local limits, mixed localization, and covariance stability.

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1 Status and provenance

This source is a mechanical extraction from the frozen 139-page Luce manuscript at repository commit `6178a99a8f2ee7b6416edd026e7780b9b8fa9fa0`. The mathematical blocks between the archival sentinels below are copied byte-for-byte from the frozen source. Only the standalone preamble, abstract, section wrappers, provenance notice, and cited bibliography entries have been added.

The extraction preserves every theorem hypothesis, identity, limiting statement, topology, quantifier, and proof step in the selected theorem package. It does not audit, repair, strengthen, weaken, or modernize the mathematics. Its status remains a working mechanical extraction until a separate adversarial audit is completed.

2 Exponential and Gumbel input

2.1 The exponential race

Let $\theta_1, \dots, \theta_n > 0$, and let $X_i \sim \mathrm{Exp}(\theta_i)$ be independent. For distinct labels $i_1, \dots, i_n$,

$$\mathbb{P}(X_{i_1} < \cdots < X_{i_n}) = \prod_{r=1}^{n} \frac{\theta_{i_r}}{\theta_{i_r} + \theta_{i_{r+1}} + \cdots + \theta_{i_n}}.$$

This is the standard exponential race identity and follows from the memoryless property.

2.2 Gumbel priorities

A standard Gumbel random variable has distribution

$$\mathbb{P}(G \le y) = \exp(-e^{-y}), \qquad y \in \mathbb{R}.$$

If $E_i \sim \mathrm{Exp}(1)$ are independent, then $-\log E_i$ are independent standard Gumbels. Since

$$X_i \overset{d}{=} E_i/\theta_i,$$

the exponential race may equivalently be generated by the shifted priorities

$$Y_i = -\log X_i = \log \theta_i + G_i, \qquad 1 \le i \le n,$$

ordered in decreasing order. Thus

$$\sigma(i) = 1 + \#\{j : Y_j > Y_i\}.$$

For later reference we record the exact threshold formulas. The density and upper-tail function of Y_i are

$$f_i(y) = \theta_i e^{-y} \exp(-\theta_i e^{-y}), \qquad \mathbb{P}(Y_i > y) = 1 - \exp(-\theta_i e^{-y}).$$

Lemma 2.1 (Gumbel threshold identities)**.** *For a single specified label p,*

$$\mathbb{P}(\sigma(p) = q) = \int_{-\infty}^{\infty} f_p(y)\, \mathbb{P}\left(\sum_{j \ne p} \mathbf{1}_{\{Y_j > y\}} = q - 1 \right) dy.$$

More generally, let $p_1, \dots, p_\ell$ be distinct labels and let $q_1 < \cdots < q_\ell$ be distinct target positions. Put $P = \{p_1, \dots, p_\ell\}$ and

$$S_a(y) = \sum_{j \notin P} \mathbf{1}_{\{Y_j > y_a\}}, \qquad 1 \le a \le \ell.$$

Then

$$\begin{aligned} &\mathbb{P}(\sigma(p_a) = q_a,\ 1 \le a \le \ell) \\ &\quad = \int_{y_1 > \cdots > y_\ell} \prod_{a=1}^{\ell} f_{p_a}(y_a)\, \mathbb{P}(S_a(y) = q_a - a,\ 1 \le a \le \ell)\, dy_1 \cdots dy_\ell. \end{aligned}$$

Proof. Condition on the specified priorities. For one label, the event $\sigma(p) = q$ is exactly the event that precisely $q - 1$ nonspecified priorities exceed the observed value of Y_p. This gives the first identity.

For the multi-point identity, the ordering of the target positions forces the specified priorities to satisfy $Y_{p_1} > \cdots > Y_{p_\ell}$. Conditional on $Y_{p_a} = y_a$ in this chamber, exactly $a - 1$ specified priorities are above y_a. Hence $\sigma(p_a) = q_a$ is equivalent to $S_a(y) = q_a - a$, simultaneously for $1 \le a \le \ell$. Integrating against the independent specified-priority densities gives the formula. □

2.3 Analytic input

We use standard estimates for sums of independent Bernoulli variables. They are stated explicitly to make clear what is invoked.

Lemma 2.2 (Point probability and Bernstein bounds)**.** *Let $Y = \sum_{j=1}^{N} \xi_j$, where the ξ_j are independent, not necessarily identically distributed, Bernoulli variables, and set $\mu = \mathbb{E}Y$, $v = \operatorname{Var} Y$. There is an absolute constant C such that*

$$\sup_m \mathbb{P}(Y = m) \le \frac{C}{\sqrt{1+v}}.$$

Moreover, there is an absolute constant $c > 0$ such that, for every $u \ge 0$,

$$\mathbb{P}(|Y - \mu| \ge u) \le 2 \exp\left[-c \min\left(\frac{u^2}{v+1}, u\right)\right].$$

Proof. The point bound is the classical concentration-function estimate for sums of Bernoulli variables. It follows from Fourier inversion and the standard inequality

$$|1 - p + pe^{it}| \le \exp\left[-\frac{2}{\pi^2} p(1-p) \operatorname{dist}(t, 2\pi\mathbb{Z})^2\right],$$

where $\operatorname{dist}(t, 2\pi\mathbb{Z})$ denotes distance from t to the nearest multiple of 2π. Bernstein's inequality gives the second bound for bounded independent summands. See Petrov [1, Chapter VII] for both estimates. □

Lemma 2.3 (Uniform lattice local CLT with a tilted local bound)**.** *Fix a dimension d and a support bound $B < \infty$. Let*

$$Y_{n,m} = \sum_{j=1}^{N_n} \zeta_{n,m,j}, \qquad m \in \mathcal{M}_n,$$

be sums of independent $\mathbb{Z}^d$-valued random vectors satisfying $\|\zeta_{n,m,j}\| \le B$. Write

$$\mu_{n,m} = \mathbb{E}Y_{n,m}, \qquad \Sigma_{n,m} = \operatorname{Var}(Y_{n,m}),$$

and suppose that, uniformly in m,

$$c s_n I_d \le \Sigma_{n,m} \le C s_n I_d, \qquad s_n \longrightarrow \infty.$$

For $\lambda \in \mathbb{R}^d$, let $\mathbb{P}_\lambda$ denote the product exponential tilt

$$\mathbb{P}_\lambda(\zeta_{n,m,j} = x) = \frac{e^{\lambda \cdot x} \mathbb{P}(\zeta_{n,m,j} = x)}{\mathbb{E} e^{\lambda \cdot \zeta_{n,m,j}}}.$$

Assume that there is $\lambda_0 > 0$ for which the following two properties hold uniformly for $\|\lambda\| \le \lambda_0$:

(i) the tilted covariance matrices satisfy

$$c_0 s_n I_d \le \operatorname{Var}_\lambda(Y_{n,m}) \le C_0 s_n I_d;$$

(ii) for every $\varepsilon > 0$ there are $\kappa_\varepsilon > 0$ and n_ε such that

$$\left| \prod_{j=1}^{N_n} \mathbb{E}_\lambda e^{iu\cdot\zeta_{n,m,j}} \right| \le e^{-\kappa_\varepsilon s_n}$$

whenever $n \ge n_\varepsilon$ and $u \in [-\pi,\pi]^d$ has distance at least ε from $2\pi\mathbb{Z}^d$.

Then, uniformly in m and $z \in \mathbb{Z}^d$,

$$\mathbb{P}(Y_{n,m} = z) = (2\pi)^{-d/2}(\det \Sigma_{n,m})^{-1/2} \times \exp\left[-\frac{1}{2}(z-\mu_{n,m})^T \Sigma_{n,m}^{-1}(z-\mu_{n,m})\right] + o(s_n^{-d/2}).$$

Moreover,

$$\sup_z \mathbb{P}(Y_{n,m} = z) \le C_1 s_n^{-d/2}.$$

For every fixed $K < \infty$ and $\gamma < 1/2$, there are $c_1, C_2 > 0$ such that, uniformly when

$$\|z - \mu_{n,m}\| \le K s_n^{1/2+\gamma},$$

one has

$$\mathbb{P}(Y_{n,m} = z) \le C_2 s_n^{-d/2} \exp\left[-c_1 \frac{\|z-\mu_{n,m}\|^2}{s_n}\right].$$

All constants depend only on the fixed dimension, the support bound, and the uniform constants in the hypotheses.

Proof. We suppress n, m and write Y, μ, Σ, s. Let $X_j = \zeta_j - \mathbb{E}\zeta_j$ and

$$\Psi(u) = \mathbb{E}e^{iu\cdot(Y-\mu)}.$$

Since $\|X_j\| \le 2B$ and $\sum_j \mathbb{E}\|X_j\|^2 = \operatorname{tr}\Sigma = O(s)$,

$$\sum_j \mathbb{E}\|X_j\|^3 = O(s).$$

Consequently, uniformly for $\|u\|$ sufficiently small,

$$\log \Psi(u) = -\frac{1}{2}u^T\Sigma u + O(s\|u\|^3). \tag{2.1}$$

A symmetrization argument also gives, on a fixed neighborhood of the origin,

$$|\Psi(u)| \le e^{-cs\|u\|^2}. \tag{2.2}$$

Indeed, if X_j' is an independent copy of X_j, then $1 - |\mathbb{E}e^{iu\cdot X_j}|^2 = \mathbb{E}[1 - \cos(u\cdot(X_j - X_j'))]$; for small u the right-hand side is bounded below by a constant multiple of $\operatorname{Var}(u\cdot X_j)$, and multiplication over j yields (2.2).

Fourier inversion on $[-\pi,\pi]^d$ now proves the local expansion. Choose $L_s \to \infty$ with $L_s^3/\sqrt{s} \to 0$ and split the integral into

$$\|u\| \le L_s/\sqrt{s}, \qquad L_s/\sqrt{s} < \|u\| \le \varepsilon, \qquad \|u\| > \varepsilon.$$

On the first region, (2.1), followed by the change of variables $v = \Sigma^{1/2}u$, gives the Gaussian Fourier integral with an $o(s^{-d/2})$ error. The estimate (2.2) makes the second region $o(s^{-d/2})$, and tilted aperiodicity with $\lambda = 0$ makes the third region exponentially small. The oscillatory factor $e^{-iu\cdot(z-\mu)}$ has modulus one, so every error is uniform in z. Taking absolute values in the same inversion formula proves the concentration bound. This is the usual fixed-dimensional Fourier proof of the lattice local central limit theorem; compare Petrov [1, Chapter VII].

It remains to prove the Gaussian local upper bound. Let $h = z - \mu$ and define the centered cumulant generating function

$$K(\lambda) = \log \mathbb{E}e^{\lambda\cdot(Y-\mu)}.$$

For $\|\lambda\| \le \lambda_0$, its Hessian is $\nabla^2 K(\lambda) = \mathrm{Var}_\lambda(Y)$ and is therefore between $c_0 s I_d$ and $C_0 s I_d$. If $\|h\| = o(s)$, strong convexity shows that the minimum of $K(\lambda) - h\cdot\lambda$ over the ball $\|\lambda\| \le \lambda_0/2$ is attained in its interior for all large n. Thus there is a unique λ_h in that ball satisfying

$$\nabla K(\lambda_h) = h, \qquad \|\lambda_h\| \le C\|h\|/s.$$

The associated Legendre transform $I(h) = \lambda_h \cdot h - K(\lambda_h)$ satisfies

$$I(h) \ge c\|h\|^2/s.$$

For example, the last inequality follows by testing the variational formula $I(h) = \sup_\lambda\{\lambda\cdot h - K(\lambda)\}$ at a sufficiently small multiple of h/s and using the tilted covariance upper bound.

Under $\mathbb{P}_{\lambda_h}$, the mean of Y is z. Repeating the absolute-value Fourier estimate above, now using the two tilted hypotheses, gives

$$\sup_w \mathbb{P}_{\lambda_h}(Y = w) \le Cs^{-d/2}.$$

The change-of-measure identity therefore yields

$$\mathbb{P}(Y = z) = e^{-I(h)}\mathbb{P}_{\lambda_h}(Y = z) \le Cs^{-d/2}e^{-c\|h\|^2/s}.$$

The stated displacement range is $o(s)$ because $\gamma < 1/2$, completing the proof. □

3 Shared anisotropic local-limit toolkit

Lemma 3.1 (Coordinatewise Fourier concentration)**.** *Let*

$$W = \sum_{\nu\in\mathcal{I}} W_\nu$$

be a sum of independent $\mathbb{Z}^r$-valued random vectors, each supported in a fixed cube $[-M_0, M_0]^r$. Let $A_1, \dots, A_r \ge 1$. Suppose that, for every coordinate j, there is a set $\mathcal{I}_j \subset \mathcal{I}$ with $|\mathcal{I}_j| \ge c_0 A_j$, and that the sets $\mathcal{I}_1, \dots, \mathcal{I}_r$ are disjoint. Suppose also that, for every $\nu \in \mathcal{I}_j$, the law of W_ν gives mass at least c_0 to two atoms whose difference is e_j, the j-th coordinate vector. Then there are constants $c, C > 0$, depending only on r, M_0, c_0, such that the characteristic function $\phi_W(u) = \mathbb{E}e^{iu\cdot W}$ satisfies

$$|\phi_W(u)| \le \exp\left[-c\sum_{j=1}^r A_j \,\mathrm{dist}(u_j, 2\pi\mathbb{Z})^2\right], \qquad u \in [-\pi, \pi]^r.$$

Consequently,

$$\sup_z \mathbb{P}(W = z) \le C \prod_{j=1}^{r} A_j^{-1/2}.$$

The same conclusions hold uniformly after any exponential tilt

$$\mathbb{P}_\lambda(W_\nu = x) = \frac{e^{\lambda \cdot x}\mathbb{P}(W_\nu = x)}{\mathbb{E}e^{\lambda \cdot W_\nu}}, \qquad \|\lambda\|_\infty \le \Lambda,$$

with constants depending additionally on Λ.

Proof. For a single summand Y, write $p_x = \mathbb{P}(Y = x)$. If two atoms x and $x + e_j$ both have mass at least c_0, then

$$\begin{aligned} 1 - |\mathbb{E}e^{iu\cdot Y}|^2 &= \frac{1}{2}\sum_{y,z} p_y p_z |e^{iu\cdot y} - e^{iu\cdot z}|^2 \\ &\ge c_0^2 |1 - e^{iu_j}|^2 \ge c\,\operatorname{dist}(u_j, 2\pi\mathbb{Z})^2. \end{aligned}$$

After decreasing c, this gives

$$|\mathbb{E}e^{iu\cdot Y}| \le \exp\{-c\operatorname{dist}(u_j, 2\pi\mathbb{Z})^2\}.$$

Multiplying this estimate over the disjoint blocks $\mathcal{I}_j$ gives the displayed characteristic-function bound. Fourier inversion then gives

$$\begin{aligned} \mathbb{P}(W = z) &\le (2\pi)^{-r} \int_{[-\pi,\pi]^r} \exp\left[-c\sum_{j=1}^{r} A_j \operatorname{dist}(u_j, 2\pi\mathbb{Z})^2\right] du \\ &\le C \prod_{j=1}^{r} A_j^{-1/2}. \end{aligned}$$

Under a tilt with $\|\lambda\|_\infty \le \Lambda$, all probabilities of atoms in the fixed support are multiplied by factors bounded between positive constants depending only on M_0, r, Λ. Hence the two distinguished atom probabilities remain bounded below, with a possibly smaller constant, and the same proof applies. □

Lemma 3.2 (Weighted chain comparison)**.** *Fix* $r \ge 1$ *and* $\eta > 0$. *Let* $0 < A_1 < \cdots < A_r$ *satisfy* $A_{j+1}/A_j \ge 1 + \eta$, *and put* $u_{r+1} = 0$. *Then*

$$\sum_{j=1}^{r} A_j u_j^2 \le C_{r,\eta} \sum_{j=1}^{r} A_j (u_j - u_{j+1})^2.$$

Equivalently, if $d_j = H_j - H_{j-1}$ *with* $H_0 = 0$, *then*

$$\sum_{j=1}^{r} \frac{d_j^2}{A_j} \asymp_{r,\eta} \sum_{j=1}^{r} \frac{H_j^2}{A_j}.$$

Proof. Write $\Delta_j = u_j - u_{j+1}$. Then $u_j = \sum_{k=j}^{r} \Delta_k$. By Cauchy's inequality with the geometrically decaying weights $A_j/A_k \le (1+\eta)^{-(k-j)}$,

$$A_j u_j^2 \le C_{r,\eta} \sum_{k=j}^{r} A_k \Delta_k^2.$$

Summing over j gives the first display. The second follows by applying the first inequality to the inverse triangular map and its reverse; since r is fixed and the ratios are bounded below, both triangular maps have uniformly bounded operator norms in the weighted norms. □

Lemma 3.3 (Truncated triangular comparison)**.** *Fix $r \ge 1$, $\eta > 0$, and $M < \infty$. Let $0 < A_1 < \cdots < A_r$ satisfy $A_{j+1}/A_j \ge 1+\eta$. If $d_j = H_j - H_{j-1}$, $H_0 = 0$, and $|d_j| \le M A_j$ for every j, then*

$$\sum_{j=1}^{r} \min\left\{ \frac{d_j^2}{A_j}, A_j \right\} \ge c_{r,\eta,M} \sum_{j=1}^{r} \min\left\{ \frac{H_j^2}{A_j}, A_j \right\}.$$

Proof. Write $x_j = d_j/A_j$. Then $|x_j| \le M$ and

$$\frac{H_j}{A_j} = \sum_{k \le j} \frac{A_k}{A_j} x_k, \qquad \frac{A_k}{A_j} \le (1+\eta)^{-(j-k)}.$$

Since $\min\{y^2, 1\} \le y^2$, Cauchy's inequality and the geometric bound on the ratios give, for each fixed j,

$$\begin{aligned} A_j \min\left\{ \frac{H_j^2}{A_j^2}, 1 \right\} &\le A_j \left(\sum_{k \le j} \frac{A_k}{A_j} x_k \right)^2 \\ &\le C_{r,\eta} A_j \sum_{k \le j} \frac{A_k}{A_j} x_k^2 = C_{r,\eta} \sum_{k \le j} A_k x_k^2. \end{aligned}$$

Because $|x_k| \le M$, we have $x_k^2 \le C_M \min\{x_k^2, 1\}$. Summing the preceding inequality over j and using that r is fixed gives

$$\sum_{j=1}^{r} A_j \min\left\{ \frac{H_j^2}{A_j^2}, 1 \right\} \le C_{r,\eta,M} \sum_{k=1}^{r} A_k \min\{x_k^2, 1\}.$$

This is exactly the claimed inequality after rewriting $A_j \min\{H_j^2/A_j^2, 1\} = \min\{H_j^2/A_j, A_j\}$ and $A_k \min\{x_k^2, 1\} = \min\{d_k^2/A_k, A_k\}$. □

Lemma 3.4 (Boundary-aware truncated chain comparison)**.** *Fix $r \ge 1$ and $M < \infty$. Let $0 < A_1 < \cdots < A_r$, let B_- and B_+ be disjoint subsets of $\{1, \dots, r\}$, and put*

$$G = \{1, \dots, r\} \setminus (B_- \cup B_+).$$

Let $H_0 = 0$, $d_j = H_j - H_{j-1}$, and

$$\phi_j(x) = \min\left\{ \frac{x^2}{A_j}, A_j \right\}.$$

Assume

$$|d_j| \le MA_j \quad (j \in G), \qquad |H_j| \le MA_j \quad (j \in B_+).$$

Then

$$\sum_{j\in G} \phi_j(d_j) + \sum_{j\in B_-} |H_j| + \sum_{j\in B_+} A_j \ge c_{r,M} \sum_{j\in G} \phi_j(H_j).$$

Proof. Decompose G into maximal consecutive runs. Let $\{\ell, \ell+1, \dots, u\}$ be one such run and put $b = \ell - 1$. For every j in the run,

$$H_j = H_b + \sum_{k=\ell}^{j} d_k,$$

where $H_b = H_0 = 0$ if $\ell = 1$. For a fixed positive number A and at most $r+1$ real numbers x_i,

$$\min\left\{\frac{(\sum_i x_i)^2}{A}, A\right\} \le C_r \sum_i \min\left\{\frac{x_i^2}{A}, A\right\}. \tag{7.1}$$

Indeed, if one term on the right equals A, the claim is immediate; otherwise it follows from Cauchy's inequality.

If $k \le j$ belongs to the run, then $A_k \le A_j$ and $|d_k| \le MA_k$. Hence

$$\phi_j(d_k) \le C_M \phi_k(d_k). \tag{7.2}$$

To see this, the assertion is immediate when $|d_k| \le A_k$; when $|d_k| > A_k$, one has $\phi_k(d_k) = A_k$ and $\phi_j(d_k) \le d_k^2/A_j \le M^2 A_k$.

It remains to control the left boundary. If $b \in B_-$, then

$$\phi_j(H_b) \le |H_b|.$$

If $b \in B_+$, then $|H_b| \le MA_b$ and $A_b \le A_j$, so

$$\phi_j(H_b) \le M^2 A_b.$$

Combining these estimates with (7.1) and (7.2), and then summing first over j in the run and finally over the finitely many runs, proves the lemma. □

Lemma 3.5 (Edge Fourier concentration)**.** *Let $e_1, \dots, e_r$ be the coordinate vectors in $\mathbb{Z}^r$ and put $e_{r+1} = 0$. Let*

$$W = \sum_{\nu\in\mathcal{I}} W_\nu$$

be a sum of independent random vectors supported on $\{e_1, \dots, e_r, 0\}$. Let $A_1, \dots, A_r \ge 1$ satisfy $A_{j+1}/A_j \ge 1+\eta$ for $j < r$. Suppose that, for every edge j, there is a set $\mathcal{I}_j \subset \mathcal{I}$ with $|\mathcal{I}_j| \ge c_0 A_j$, the sets $\mathcal{I}_j$ are disjoint, and for every $\nu \in \mathcal{I}_j$ the two atoms e_j and e_{j+1} each have probability at least c_0. Then

$$|\mathbb{E}e^{iu\cdot W}| \le \exp\left[-c\sum_{j=1}^{r} A_j \operatorname{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2\right], \qquad u_{r+1} = 0,$$

and consequently

$$\sup_z \mathbb{P}(W = z) \le C \prod_{j=1}^{r} A_j^{-1/2}.$$

The same conclusions hold uniformly after any exponential tilt $\|\lambda\|_\infty \le \Lambda$*, with constants depending additionally on* Λ*.*

Proof. If a summand Y gives mass at least c_0 to e_j and to e_{j+1}, then

$$1 - |\mathbb{E}e^{iu\cdot Y}|^2 \ge c_0^2 |e^{iu_j} - e^{iu_{j+1}}|^2 \ge c \operatorname{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2.$$

As in Lemma 3.1, this implies

$$|\mathbb{E}e^{iu\cdot Y}| \le \exp\{-c \operatorname{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2\}.$$

Multiplying over the disjoint edge blocks gives the displayed Fourier bound. For the concentration estimate, Fourier inversion and Lemma 3.2 give

$$\begin{aligned}\mathbb{P}(W = z) &\le (2\pi)^{-r} \int_{[-\pi,\pi]^r} \exp\left[-c \sum_{j=1}^{r} A_j \operatorname{dist}(u_j - u_{j+1}, 2\pi\mathbb{Z})^2\right] du \\ &\le C \prod_{j=1}^{r} A_j^{-1/2}.\end{aligned}$$

Bounded tilts only multiply the finitely many atom probabilities by bounded factors, so the edge-block lower bounds persist after changing constants. □

Lemma 3.6 (Forward-edge projected concentration)**.** *Let* $G \ne \emptyset$ *be a subset of* $\{1, \dots, r\}$ *and let* $0 < A_1 < \cdots < A_r$ *satisfy* $A_{j+1}/A_j \ge 1 + \eta$*. Let* W *be a sum of independent random vectors supported on* $\{e_j : j \in G\} \cup \{0\}$ *after projection onto the coordinates in* G*. Suppose that for each* $j \in G$ *there is an index* $\kappa(j) \in (G \cap \{j+1, \dots, r\}) \cup \{0\}$ *and a set of at least* $c_0 A_j$ *summands, disjoint as* j *varies, for which the two projected atoms* e_j *and* $e_{\kappa(j)}$ *have probabilities at least* c_0*; here* $e_0 = 0$*. Then, uniformly in* z_G*,*

$$\mathbb{P}(W_G = z_G) \le C \prod_{j \in G} A_j^{-1/2}.$$

The same estimate holds after any exponential tilt whose coordinates are bounded in absolute value by a fixed constant.

Proof. For a summand in the block assigned to j, the same two-atom argument used in Lemma 3.5 gives the factor

$$\exp\{-c \operatorname{dist}(u_j - u_{\kappa(j)}, 2\pi\mathbb{Z})^2\}, \qquad u_0 = 0.$$

Multiplying over the disjoint blocks gives

$$|\phi(u)| \le \exp\left[-c \sum_{j \in G} A_j \operatorname{dist}(u_j - u_{\kappa(j)}, 2\pi\mathbb{Z})^2\right].$$

Because every directed edge goes from j to a larger index, or to the anchor 0, each path reaches 0 after at most r steps. Along such a path the weights only increase. Therefore, for $u_0 = 0$,

$$\sum_{j\in G} A_j u_j^2 \le C_{r,\eta} \sum_{j\in G} A_j (u_j - u_{\kappa(j)})^2.$$

The torus version follows from the same path argument. For each j, choose integer multiples of 2π successively along the directed path $P(j)$ from j to the anchor 0. The triangle inequality on the circle gives

$$\operatorname{dist}(u_j, 2\pi\mathbb{Z}) \le \sum_{\ell\in P(j)} \operatorname{dist}(u_\ell - u_{\kappa(\ell)}, 2\pi\mathbb{Z}).$$

Cauchy's inequality, together with monotonicity of the weights along this path, gives the displayed comparison with distances to $2\pi\mathbb{Z}$. Fourier inversion then gives

$$\mathbb{P}(W_G = z_G) \le C \int_{[-\pi,\pi]^G} \exp\left[-c \sum_{j\in G} A_j u_j^2\right] du_G \le C \prod_{j\in G} A_j^{-1/2}.$$

Bounded tilts distort all atoms in the finite support by bounded factors, so the two-atom lower bounds persist after changing constants. □

Lemma 3.7 (Terminal interval edge blocks)**.** *Fix $r \ge 1$, $0 < c_* < C_* < \infty$, and $c_0 > 0$. In the terminal interval notation above, fix $j \in \{1, \dots, r\}$ and suppose*

$$c_* \le s_j = a_j t_j \le C_*, \qquad m_j(t) = \sum_{b\notin A} \mathbb{P}(T_b \in I_j) \ge c_0 a_j.$$

Then there are constants $0 < \alpha < \beta < \infty$ and $c > 0$, depending only on r, c_*, C_*, and c_0, and a set of at least ca_j rates $b \in [\alpha a_j, \beta a_j] \setminus A$ such that

$$\mathbb{P}(T_b \in I_j) \ge c, \qquad \mathbb{P}(T_b \le t_j) \ge c.$$

Consequently, after projecting any chosen subset of the interval coordinates larger than j either to their own coordinates or to the anchor 0, a further subblock of size at least ca_j gives two projected atoms e_j and $e_{\kappa(j)}$, where $\kappa(j) > j$ is a retained coordinate or $\kappa(j) = 0$, each with probability at least c. The conclusion remains valid under every interval-coordinate exponential tilt whose coordinates are bounded by a fixed constant, after changing c.

Proof. Choose $\alpha > 0$ small and $\beta < \infty$ large. The contribution to $m_j(t)$ from $b < \alpha a_j$ is at most $\alpha a_j + 1$. Since $t_j \ge c_*/a_j$, the contribution from $b > \beta a_j$ is at most

$$\sum_{b>\beta a_j} e^{-bt_j} \le C a_j e^{-c_*\beta}.$$

After first decreasing α and then increasing β, at least a fixed positive multiple of a_j of the total interval mass comes from $[\alpha a_j, \beta a_j]$. This block contains $O(a_j)$ rates, so at least ca_j of them satisfy $\mathbb{P}(T_b \in I_j) \ge c$. For every such rate,

$$bt_j \ge \alpha c_*, \qquad \mathbb{P}(T_b \le t_j) = 1 - e^{-bt_j} \ge 1 - e^{-\alpha c_*} > 0.$$

The event $\{T_b \le t_j\}$ is the disjoint union of the later categories $I_{j+1}, \dots, I_{r+1}$. For each selected rate, one of these finitely many categories has probability bounded below. A second pigeonhole argument produces a common later category on a subblock of size ca_j. Projection maps that category either to a retained atom $e_{\kappa(j)}$ or to the anchor 0. Removing the fixed set A changes the block size by only $O_r(1)$. A bounded exponential tilt changes every atom probability in the finite categorical support by a bounded multiplicative factor, which proves the final assertion. □

Lemma 3.8 (Projected good–bad interval bound)**.** *Fix $r \ge 1$, $\eta > 0$, and a compact scaled chamber $K \subset (0,\infty)^r$. In the terminal interval notation above, suppose $s = (a_1 t_1, \dots, a_r t_r) \in K$ and $t_1 > \cdots > t_r$. Suppose the deterministic targets satisfy*

$$c_0 a_j \le y_j \le C_0 a_j, \qquad 1 \le j \le r.$$

Then, uniformly for such t and y,

$$\mathbb{P}(Y(t) = y) \le C \prod_{j=1}^{r} a_j^{-1/2} \exp\left[-c \sum_{j=1}^{r} \min\left\{ \frac{(y_j - m_j(t))^2}{a_j}, a_j \right\} \right].$$

The constants depend only on r, η, K, c_0, and C_0.

Proof. Write $d_j = y_j - m_j(t)$. Since $s \in K$, finite-geometric estimates give $m_j(t) = O_K(a_j)$ and hence $|d_j| \le M_K a_j$. Fix a small $\varepsilon > 0$. Let

$$G = \{j : |d_j| \le \varepsilon a_j\}, \qquad B = \{1, \dots, r\} \setminus G.$$

For $j \in G$ the target lower bound gives $m_j(t) \ge (c_0 - \varepsilon) a_j$; choose $\varepsilon < c_0/2$, so Lemma 3.7 applies to every good coordinate.

Use one combined exponential tilt. Put

$$\lambda_j = \begin{cases} \delta d_j / a_j, & j \in G, \\ \delta \operatorname{sgn}(d_j), & j \in B, \end{cases}$$

where $\delta > 0$ is a sufficiently small fixed constant. Let $K_t(\lambda) = \log \mathbb{E} \exp\{\lambda \cdot (Y(t) - m(t))\}$. For every bounded tilt ξ, the categorical support and the finite-geometric bound $\operatorname{Var}_\xi Y_j(t) \le C a_j$ give

$$\operatorname{Var}_\xi(v \cdot Y(t)) \le C \sum_{j=1}^{r} a_j v_j^2.$$

Therefore

$$K_t(\lambda) \le C \sum_{j=1}^{r} a_j \lambda_j^2.$$

On the other hand,

$$\lambda \cdot d \ge \delta \sum_{j \in G} \frac{d_j^2}{a_j} + \delta\varepsilon \sum_{j \in B} a_j.$$

Choosing δ small enough yields

$$-\lambda \cdot d + K_t(\lambda) \le -c \sum_{j \in G} \frac{d_j^2}{a_j} - c \sum_{j \in B} a_j.$$

Under this bounded tilt all categorical probabilities are changed by bounded multiplicative factors. Project the tilted array onto the good coordinates and ignore the bad-coordinate constraints. For each $j \in G$, Lemma 3.7 gives a candidate block B_j of at least $c_0 a_j$ rates with two projected atoms e_j and $e_{\kappa(j)}$, where $\kappa(j) > j$ is another good coordinate or $\kappa(j) = 0$. We now extract disjoint subblocks rigorously. Process the indices of G in increasing order. Since $a_{i+1}/a_i \geq 1 + \eta$,

$$\sum_{i<j} a_i \leq C_{r,\eta} a_j.$$

Choose $c' > 0$ so small that $c'(1 + C_{r,\eta}) \leq c_0$. After subblocks of sizes $c'a_i$ have been chosen for all earlier good indices $i < j$, at most $c' \sum_{i<j} a_i \leq c' C_{r,\eta} a_j$ elements of B_j have been used. At least $c'a_j$ unused elements of B_j therefore remain. Continuing greedily produces disjoint subblocks of size at least $c'a_j$ for every $j \in G$. Lemma 3.6 gives

$$\mathbb{P}_\lambda(Y_G(t) = y_G) \leq C \prod_{j \in G} a_j^{-1/2}.$$

Consequently

$$\begin{aligned}\mathbb{P}(Y(t) = y) &\leq \exp\{-\lambda \cdot d + K_t(\lambda)\} \mathbb{P}_\lambda(Y_G(t) = y_G) \\ &\leq C \prod_{j \in G} a_j^{-1/2} \exp\left[-c \sum_{j \in G} \frac{d_j^2}{a_j} - c \sum_{j \in B} a_j \right].\end{aligned}$$

For $j \in B$, the factor e^{-ca_j} absorbs the missing local factor: $e^{-ca_j} \leq C a_j^{-1/2} e^{-c'a_j}$. Since for $j \in G$ we have $|d_j| \leq \varepsilon a_j$, this is the desired bound, after changing constants.

□

Lemma 3.9 (Actual-mean cumulative Chernoff bounds)**.** *Fix $r \geq 1$. Let $B \subset \{1, \dots, r\}$, and put*

$$\mu_j = \mathbb{E} Z_j(t), \qquad j \in B.$$

There are constants $\delta_0, C > 0$, depending only on r, such that, for every $|\theta| \leq \delta_0$,

$$\log \mathbb{E} \exp\left\{ \theta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \leq C\theta^2 \sum_{j \in B} \mu_j.$$

After decreasing δ_0, if necessary, fix $0 < \delta \leq \delta_0$ and $C_0 < \infty$. There are $c = c(\delta, r, C_0) > 0$ and $L_0 = L_0(\delta, r, C_0)$ such that, whenever $L \geq L_0$, both of the following assertions hold.

If $z_j \leq C_0 L^{-1} \mu_j$ for every $j \in B$, then

$$-\sum_{j \in B} (-\delta)(z_j - \mu_j) + \log \mathbb{E} \exp\left\{ -\delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \leq -c \sum_{j \in B} \mu_j.$$

If $\mu_j \leq C_0 L^{-1} z_j$ for every $j \in B$, then

$$-\sum_{j \in B} \delta(z_j - \mu_j) + \log \mathbb{E} \exp\left\{ \delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \leq -c \sum_{j \in B} z_j.$$

In particular, the first exponent is at most $-cL \sum_{j \in B} a_j$ if $\mu_j \geq c_0 L a_j$, while the second is at most $-c \sum_{j \in B} a_j$ if $z_j \geq c_0 a_j$.

Proof. For a single rate $b \notin A$, define

$$R_b = \sum_{j \in B} \mathbf{1}_{\{T_b > t_j\}}, \qquad m_b = \mathbb{E} R_b.$$

Then $0 \le R_b \le r$, and

$$\sum_{b \notin A} m_b = \sum_{j \in B} \mu_j.$$

For $|\theta| \le 1$, let

$$K_b(\theta) = \log \mathbb{E} e^{\theta(R_b - m_b)}.$$

Then $K_b(0) = K_b'(0) = 0$, and

$$K_b''(\theta) = \mathrm{Var}_\theta(R_b) \le \mathbb{E}_\theta R_b^2 \le r\, \mathbb{E}_\theta R_b.$$

Since $0 \le R_b \le r$, a bounded tilt changes the expectation of the nonnegative variable R_b by at most a factor depending only on r:

$$\mathbb{E}_\theta R_b = \frac{\mathbb{E} R_b e^{\theta R_b}}{\mathbb{E} e^{\theta R_b}} \le e^{2r} m_b, \qquad |\theta| \le 1.$$

Therefore $K_b''(\theta) \le C_r m_b$ for $|\theta| \le 1$, and Taylor's formula gives $K_b(\theta) \le C_r \theta^2 m_b$. Summing over $b \notin A$ proves the first displayed inequality for both signs of θ.

For the lower-tail assertion, use the preceding bound and $z_j \le C_0 L^{-1} \mu_j$:

$$\begin{aligned} &- \sum_{j \in B} (-\delta)(z_j - \mu_j) + \log \mathbb{E} \exp \left\{ -\delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \\ &\qquad \le -\delta \left(1 - \frac{C_0}{L} \right) \sum_{j \in B} \mu_j + C\delta^2 \sum_{j \in B} \mu_j. \end{aligned}$$

Decrease δ_0, if necessary, so that $C\delta_0 \le 1/4$, and then choose L_0 so that $C_0/L \le 1/4$. The last display is at most $-\delta \sum_{j \in B} \mu_j/2$.

For the upper-tail assertion, the same cumulant bound and $\mu_j \le C_0 L^{-1} z_j$ give

$$\begin{aligned} &- \sum_{j \in B} \delta(z_j - \mu_j) + \log \mathbb{E} \exp \left\{ \delta \sum_{j \in B} (Z_j(t) - \mu_j) \right\} \\ &\qquad \le -\delta \sum_{j \in B} z_j + (\delta + C\delta^2) \sum_{j \in B} \mu_j \\ &\qquad \le -\delta \left(1 - \frac{2C_0}{L} \right) \sum_{j \in B} z_j, \end{aligned}$$

after decreasing δ_0 once more. Enlarging L_0 makes this at most $-\delta \sum_{j \in B} z_j/2$. The two final scale bounds follow immediately from the corresponding assumptions. □

Lemma 3.10 (Mixed cumulative–interval localization)**.** *Fix $r \ge 1$ and $\eta > 0$. There is $L_0 = L_0(r, \eta)$ such that the following holds for every $L \ge L_0$. In the terminal interval notation, let $s_j = a_j t_j$ and suppose $s \notin [L^{-1}, L]^r$. Put*

$$B_-(s) = \{j : s_j < L^{-1}\}, \qquad B_+(s) = \{j : s_j > L\}, \qquad G(s) = \{1, \dots, r\} \setminus (B_-(s) \cup B_+(s)).$$

On every region on which these three sets are fixed,

$$\mathbb{P}(Y(t)=y) \le C_L \prod_{j=1}^{r} a_j^{-1/2} \exp\left[-c\sum_{j\in B_-}|H_j(t)| - c(\log L)\sum_{j\in B_+} a_j\right] \times \exp\left[-c_L \sum_{j\in G(s)} \min\left\{\frac{H_j(t)^2}{a_j}, a_j\right\}\right]. \tag{7.3}$$

Here $c>0$ is independent of L, while $C_L<\infty$ and $c_L>0$ depend only on L, r, and η. Consequently,

$$\mathbb{P}(Y(t)=y) \le C_L \prod_{j=1}^{r} a_j^{-1/2} \exp\left[-cL\sum_{j\in B_-} a_j - c(\log L)\sum_{j\in B_+} a_j\right] \times \exp\left[-c_L \sum_{j\in G(s)} a_j \min\{(s_j-s_j^*)^2, 1\}\right]. \tag{7.4}$$

Proof. Throughout the proof the offender sets are fixed. Write $B_- = B_-(s)$, $B_+ = B_+(s)$, and $G = G(s)$. Finite-geometric estimates give, uniformly in the admissible terminal range,

$$\begin{aligned} j\in B_-: &\quad \mu_j(t_j) \ge cLa_j, \qquad H_j(t) = z_j - \mu_j(t_j) < 0, \\ j\in B_+: &\quad \mu_j(t_j) \le Ca_j/L, \quad 0 < H_j(t) \le z_j \le a_j, \end{aligned} \tag{7.5}$$

for L sufficiently large. In particular, $|H_j(t)| \ge cLa_j$ on B_- and $H_j(t) \ge ca_j$ on B_+. For $j\in G$, one has $s_j \in [L^{-1}, L]$, and therefore

$$m_j(t) \le \mu_j(t_j) \le C_L a_j, \qquad |d_j(t)| \le M_L a_j. \tag{7.6}$$

Let

$$K_t(\theta) = \log \mathbb{E}\exp\{\theta\cdot(Y(t)-m(t))\}$$

be the centered interval-coordinate cumulant. If $\gamma=(\gamma_1,\dots,\gamma_r)$ is a cumulative-coordinate vector, put

$$(\mathcal{T}\gamma)_k = \sum_{j\ge k}\gamma_j.$$

Since $Z_j = Y_1+\cdots+Y_j$ and $d_j = H_j - H_{j-1}$,

$$(\mathcal{T}\gamma)\cdot d = \gamma\cdot H. \tag{7.7}$$

Choose $\delta>0$ so small that 3δ is admissible in Lemma 3.9 and $3\delta r < 1/2$. Define

$$\gamma_j^- = -\delta \mathbf{1}_{\{j\in B_-\}}, \qquad \gamma_j^+ = \delta(\log L)\mathbf{1}_{\{j\in B_+\}}.$$

We also introduce a local interval tilt on G. Ratio separation implies that there is $c_\eta>0$ such that $y_j \ge c_\eta a_j$ for all j and all sufficiently large n. Fix $0<\varepsilon<c_\eta/4$ and put

$$G_0 = \{j\in G : |d_j| \le \varepsilon a_j\}, \qquad G_1 = G\setminus G_0.$$

For a constant $\delta_L > 0$ to be chosen below, set

$$\lambda_j = \begin{cases} \delta_L d_j / a_j, & j \in G_0, \\ \delta_L \operatorname{sgn}(d_j), & j \in G_1, \\ 0, & j \notin G. \end{cases}$$

Finally, let

$$\theta = \mathcal{T}\gamma^- + \mathcal{T}\gamma^+ + \lambda.$$

Generalized Hölder inequality gives the precise combined-cumulant estimate

$$K_t(\theta) \le \frac{1}{3} K_t(3\mathcal{T}\gamma^-) + \frac{1}{3} K_t(3\mathcal{T}\gamma^+) + \frac{1}{3} K_t(3\lambda). \tag{7.8}$$

Thus, by (7.7),

$$\begin{aligned} -\theta \cdot d + K_t(\theta) \le{}& \frac{1}{3}\{-3\gamma^- \cdot H + K_t(3\mathcal{T}\gamma^-)\} \\ &+ \frac{1}{3}\{-3\gamma^+ \cdot H + K_t(3\mathcal{T}\gamma^+)\} \\ &+ \frac{1}{3}\{-3\lambda \cdot d + K_t(3\lambda)\}. \end{aligned} \tag{7.9}$$

We estimate the three braces separately.

For B_-, (7.5) implies $z_j \le CL^{-1}\mu_j(t_j)$. Lemma 3.9, applied with parameter 3δ, gives

$$-3\gamma^- \cdot H + K_t(3\mathcal{T}\gamma^-) \le -c \sum_{j \in B_-} \mu_j(t_j) \le -c \sum_{j \in B_-} |H_j(t)|. \tag{7.10}$$

The same bound also retains the weaker loss $-cL \sum_{j \in B_-} a_j$.

For B_+, put

$$R_b^+ = \sum_{j \in B_+} \mathbf{1}_{\{T_b > t_j\}}, \qquad 0 \le R_b^+ \le r.$$

Since the summands indexed by $b \notin A$ are independent,

$$\begin{aligned} K_t(3\mathcal{T}\gamma^+) &= \sum_{b \notin A} \log \mathbb{E} \exp\{3\delta(\log L)(R_b^+ - \mathbb{E}R_b^+)\} \\ &\le \sum_{b \notin A} \log \mathbb{E} \exp\{3\delta(\log L) R_b^+\} \\ &\le L^{3\delta r} \sum_{b \notin A} \mathbb{P}(R_b^+ > 0) \\ &\le L^{3\delta r} \sum_{j \in B_+} \mu_j(t_j) \le CL^{-1/2} \sum_{j \in B_+} a_j. \end{aligned} \tag{7.11}$$

On the other hand, (7.5) gives

$$-3\gamma^+ \cdot H \le -c(\log L) \sum_{j \in B_+} a_j.$$

After increasing L_0, (7.11) is absorbed by this linear gain, and hence

$$-3\gamma^+ \cdot H + K_t(3\mathcal{T}\gamma^+) \le -c(\log L) \sum_{j \in B_+} a_j. \tag{7.12}$$

For the local tilt, let v vanish outside G. Under every bounded interval-coordinate tilt ξ, a categorical summand contributes at most one coordinate, and therefore

$$\operatorname{Var}_\xi(v \cdot Y(t)) \le \sum_{j\in G} v_j^2 \mathbb{E}_\xi Y_j(t) \le C_L \sum_{j\in G} a_j v_j^2, \tag{7.13}$$

where the last inequality follows from (7.6) and bounded distortion of atom probabilities. Taylor's formula for the cumulant gives

$$K_t(3\lambda) \le C_L \sum_{j\in G} a_j \lambda_j^2.$$

Choose $\delta_L > 0$ sufficiently small. Then

$$-3\lambda \cdot d + K_t(3\lambda) \le -c_L \sum_{j\in G} \min\left\{ \frac{d_j^2}{a_j}, a_j \right\}. \tag{7.14}$$

Indeed, on G_0 the linear term is a fixed multiple of d_j^2/a_j, whereas on G_1 it is at least a fixed multiple of a_j.

We next recover the product local factor under the single total tilt θ. For fixed L, its interval coordinates are bounded by a constant depending only on L and r. If $j \in G_0$, then

$$m_j(t) \ge y_j - \varepsilon a_j \ge (c_\eta/2) a_j.$$

Lemma 3.7, with $s_j \in [L^{-1}, L]$, supplies a candidate forward-edge block of order a_j. Process the indices of G_0 increasingly and choose small subblocks greedily. Since $a_{i+1}/a_i \ge 1+\eta$ and r is fixed,

$$\sum_{i<j} a_i \le C_{r,\eta} a_j,$$

so disjoint subblocks of size $c_{L,r,\eta} a_j$ can be retained for every $j \in G_0$. Boundedness of the total tilt preserves the two-atom lower bounds. Lemma 3.6, applied after projection onto G_0, yields

$$\mathbb{P}_\theta(Y_{G_0}(t) = y_{G_0}) \le C_L \prod_{j\in G_0} a_j^{-1/2}. \tag{7.15}$$

The exact change-of-measure identity and (7.9)–(7.15) now give

$$\begin{aligned} \mathbb{P}(Y(t) = y) \le C_L \prod_{j\in G_0} a_j^{-1/2} \exp\left[-c \sum_{j\in B_-} |H_j| - c(\log L) \sum_{j\in B_+} a_j \right] \\ \times \exp\left[-c_L \sum_{j\in G} \min\left\{ \frac{d_j^2}{a_j}, a_j \right\} \right]. \end{aligned} \tag{7.16}$$

For $j \in G_1$, the last exponent contains $-c_L a_j$; for $j \in B_-$ and $j \in B_+$, (7.5) and the preceding tail losses contain the same fixed-order exponential cost. These factors absorb all missing $a_j^{-1/2}$ terms, after changing constants.

It remains to pass from interval residuals to cumulative residuals without losing boundary terms. The assumptions of Lemma 3.4 hold with $A_j = a_j$ and $M = M_L$: (7.6) controls d_j on G,

while (7.5) gives $|H_j| \le a_j$ on B_+. Split a fixed fraction of each offender loss in (7.16) for this comparison. Since $\log L$ is large, Lemma 3.4 yields

$$\begin{aligned} c_L \sum_{j\in G} \min\left\{\frac{d_j^2}{a_j}, a_j\right\} + c \sum_{j\in B_-} |H_j| + c(\log L) \sum_{j\in B_+} a_j \\ \ge c'_L \sum_{j\in G} \min\left\{\frac{H_j^2}{a_j}, a_j\right\} + c' \sum_{j\in B_-} |H_j| + c'(\log L) \sum_{j\in B_+} a_j. \end{aligned} \tag{7.17}$$

Combining (7.16) and (7.17) proves (7.3).

Finally, on G the scaled coordinate s_j belongs to $[L^{-1}, L]$. Uniform finite-geometric derivative estimates and strict monotonicity give

$$|H_j(t)| = |\mu_j(t_j^*) - \mu_j(t_j)| \ge c_L a_j \min\{|s_j - s_j^*|, 1\}. \tag{7.18}$$

Together with (7.5), this converts (7.3) into (7.4). □

Lemma 3.11 (Anisotropic tilted local upper bound)**.** *Let $W = \sum_{\nu\in\mathcal{I}} W_\nu$ be as in Lemma 3.1. Assume that the hypotheses of Lemma 3.1 hold uniformly under all tilts $\|\lambda\|_\infty \le \Lambda$. Assume also that, for those tilted laws,*

$$\mathrm{Var}_\lambda(v \cdot W) \le C_0 \sum_{j=1}^r A_j v_j^2, \qquad v \in \mathbb{R}^r.$$

Then there are constants $C, c > 0$, depending only on the constants in the assumptions, such that for every lattice point z, with $h = z - \mathbb{E}W$,

$$\mathbb{P}(W = z) \le C \prod_{j=1}^r A_j^{-1/2} \exp\left[-c \sum_{j=1}^r \min\left\{\frac{h_j^2}{A_j}, |h_j|\right\}\right].$$

In particular, if $|h_j| \le A_j$ for every j, then the exponent may be written as $-c\sum_j h_j^2/A_j$.

Proof. Let

$$K(\lambda) = \log \mathbb{E} e^{\lambda\cdot(W - \mathbb{E}W)}.$$

For $\|\lambda\|_\infty \le \Lambda$, the covariance upper bound gives

$$K(\lambda) = \int_0^1 (1-s)\lambda^T \mathrm{Var}_{s\lambda}(W)\lambda \, ds \le C \sum_{j=1}^r A_j \lambda_j^2.$$

Choose a sufficiently small $\delta > 0$, depending only on the constants in the assumptions and with $\delta \le \Lambda$, and put

$$\lambda_j = \delta \,\mathrm{sgn}(h_j) \min\{|h_j|/A_j, 1\}.$$

By tilting and then applying Lemma 3.1 under the tilted law,

$$\begin{aligned} \mathbb{P}(W = z) &= \exp\{-\lambda\cdot h + K(\lambda)\}\mathbb{P}_\lambda(W = z) \\ &\le C \prod_{j=1}^r A_j^{-1/2} \exp\{-\lambda\cdot h + K(\lambda)\}. \end{aligned}$$

For the chosen λ,

$$\lambda \cdot h = \delta \sum_{j=1}^{r} |h_j| \min\{|h_j|/A_j, 1\},$$

while

$$K(\lambda) \le C\delta^2 \sum_{j=1}^{r} A_j \min\{h_j^2/A_j^2, 1\}.$$

Taking δ small enough makes the second display at most one half of the first, after changing constants. This proves the claim. □

4 Endpoint-regular fixed-point universality

We now promote the endpoint fixed-point prediction to a theorem. The statement is formulated in terminal rate coordinates. Put

$$\lambda_{n,a} = \theta_{n,n+1-a}, \qquad 1 \le a \le n,$$

so that terminal coordinate a is label $n+1-a$. Let

$$T_{n,a} \sim \mathrm{Exp}(\lambda_{n,a}), \qquad \tau_n(a) = 1 + \sum_{b \ne a} \mathbf{1}_{\{T_{n,b} > T_{n,a}\}} = n + 1 - \sigma_n(n+1-a).$$

Thus τ_n is conjugate to σ_n by the reversal permutation and has the same cycle counts. In Gumbel variables

$$Y_{n,a} = \log \lambda_{n,a} + G_a,$$

the fixed-point indicator is

$$I_{n,a} = \mathbf{1}\left\{ \sum_{b \ne a} \mathbf{1}_{\{Y_{n,b} < Y_{n,a}\}} = a - 1 \right\}.$$

Indeed, label $n+1-a$ is fixed exactly when there are $a-1$ labels below it in priority order. We write

$$F_n = \sum_{a=1}^{n} I_{n,a} = C_1(\sigma_n)$$

inside this section.

Multiplying all rates by a common deterministic factor adds the same constant to all priorities and does not change the Luce order. The following definition therefore fixes a convenient normalization.

Definition 4.1 (Uniform endpoint regular variation). The terminal rate array $(\lambda_{n,a})_{1 \le a \le n}$ is *uniformly endpoint regularly varying with index* $\gamma > 0$ if there exist positive constants s_n, a constant $C_0 > 0$, and a positive measurable slowly varying function L at infinity such that the normalized rates

$$\bar{\lambda}_{n,a} = s_n \lambda_{n,a}$$

satisfy the following conditions.

(ER1) For every sequence $A_n \to \infty$ with $A_n = o(n)$,

$$\sup_{1 \le a \le A_n} \left| \frac{\bar{\lambda}_{n,a}}{C_0 (a/n)^\gamma L(n/a)} - 1 \right| \longrightarrow 0.$$

(ER2) For every $\delta > 0$, there are constants $0 < c_\delta < C_\delta < \infty$ such that

$$c_\delta \le \bar{\lambda}_{n,a} \le C_\delta, \qquad \delta n \le a \le n,$$

for all sufficiently large n.

(ER3) The normalized terminal rates are nondecreasing:

$$\bar{\lambda}_{n,1} \le \bar{\lambda}_{n,2} \le \cdots \le \bar{\lambda}_{n,n}.$$

A common multiplication of all rates does not change the Luce order. We therefore replace $\lambda_{n,a}$ by $\bar{\lambda}_{n,a}$ throughout this section and suppress the bar. We use repeatedly the uniform convergence theorem and Potter bounds for slowly varying functions in the forms of Bingham, Goldie, and Teugels [2, Theorems 1.2.1 and 1.5.6].

For example, suppose $\lambda_{n,a} = f(1 - a/n)$, where

$$f(1-u) \sim C_0 u^\gamma L(1/u), \qquad u \downarrow 0,$$

where f is bounded above and below away from zero on every compact subinterval of $[0, 1)$, and where the function $u \mapsto f(1-u)$ is nondecreasing on $(0, 1]$. Then the sampled array satisfies Definition 4.1. The global monotonicity in this example is intentional: eventual endpoint monotonicity alone would not imply condition (ER3) for the full triangular array.

Lemma 4.2 (The endpoint partial-sum bound is automatic)**.** *Under Definition 4.1, there is $c_\Sigma > 0$ such that, for all sufficiently large n,*

$$\sum_{b=1}^{a} \lambda_{n,b} \ge c_\Sigma a \lambda_{n,a}, \qquad 1 \le a \le n.$$

Proof. Suppose otherwise. Then there are $n_k \to \infty$ and $a_k \in [n_k]$ such that

$$\frac{\sum_{b \le a_k} \lambda_{n_k,b}}{a_k \lambda_{n_k,a_k}} \longrightarrow 0.$$

Pass to a subsequence. If (a_k) is bounded, the ratio is at least $1/a_k$, a contradiction. Assume therefore that $a_k \to \infty$.

If $a_k/n_k \to 0$, uniform endpoint regular variation and Potter bounds give

$$\frac{1}{a_k} \sum_{b=1}^{a_k} \frac{\lambda_{n_k,b}}{\lambda_{n_k,a_k}} \longrightarrow \int_0^1 x^\gamma \, dx = \frac{1}{\gamma + 1}.$$

For completeness, on $\varepsilon a_k \le b \le a_k$ this is an ordinary Riemann sum by the uniform convergence theorem for slowly varying functions. On $b < \varepsilon a_k$, Potter's bound gives $\lambda_{n_k,b}/\lambda_{n_k,a_k} \le C(b/a_k)^{\gamma/2}$, so the omitted part is $O(\varepsilon^{1+\gamma/2})$. Letting $\varepsilon \downarrow 0$ proves the display, again a contradiction.

The remaining case has $a_k \geq \delta n_k$ for some $\delta > 0$. By monotonicity and (ER2),

$$\sum_{b=1}^{a_k} \lambda_{n_k,b} \geq \sum_{b=\lceil a_k/2 \rceil}^{a_k} \lambda_{n_k,b} \geq \frac{a_k}{2} c_{\delta/2},$$

whereas $\lambda_{n_k,a_k} \leq C_\delta$. The ratio is therefore bounded below by $c_{\delta/2}/(2C_\delta) > 0$, the final contradiction. □

Set

$$A_\gamma = \Gamma(1 + 1/\gamma), \qquad K_\gamma = A_\gamma^\gamma, \qquad c_\gamma = \gamma K_\gamma e^{-K_\gamma}.$$

For fixed $r \geq 1$ and $\eta > 0$, define the admissible family

$$\mathfrak{E}_N(r,\eta) = \left\{ (n, a_1, \ldots, a_r) : N \leq a_1 < \cdots < a_r \leq n/N, \quad \frac{a_{j+1}}{a_j} \geq 1 + \eta \right\}.$$

Here and below, a statement asserted uniformly for $1 \ll a_1 < \cdots < a_r \ll n$ means that its supremum over $\mathfrak{E}_N(r,\eta)$ tends to zero as $N \to \infty$. This convention makes all endpoint uniformity statements independent of a particular choice of intermediate cutoff sequences.

Lemma 4.3 (Weighted regularly varying terminal sums)**.** *Assume Definition 4.1. Let $a = a_n \to \infty$ with $a = o(n)$. For every fixed integer $m \geq 0$, put*

$$J_{\gamma,m}(\kappa) = \int_0^\infty x^{m\gamma} e^{-\kappa x^\gamma}\, dx = \frac{1}{\gamma}\Gamma\left(m + \frac{1}{\gamma}\right) \kappa^{-m-1/\gamma}.$$

Then, uniformly for κ in compact subintervals of $(0,\infty)$,

$$\sum_{b=1}^{n} \left(\frac{\lambda_{n,b}}{\lambda_{n,a}}\right)^m \exp\left(-\kappa \frac{\lambda_{n,b}}{\lambda_{n,a}}\right) = a J_{\gamma,m}(\kappa)(1 + o(1)). \tag{8.1}$$

The same assertion holds after deleting any fixed number of summands, uniformly in the deleted indices. In particular,

$$\sum_{b=1}^{n} \exp\left(-\kappa \frac{\lambda_{n,b}}{\lambda_{n,a}}\right) = a A_\gamma \kappa^{-1/\gamma}(1 + o(1)), \tag{8.2}$$

$$\sum_{b=1}^{n} \frac{\lambda_{n,b}}{\lambda_{n,a}} \exp\left(-\kappa \frac{\lambda_{n,b}}{\lambda_{n,a}}\right) = \frac{a A_\gamma}{\gamma} \kappa^{-1-1/\gamma}(1 + o(1)), \tag{8.3}$$

and, with $p_{n,b}(\kappa) = \exp(-\kappa \lambda_{n,b}/\lambda_{n,a})$,

$$\sum_{b=1}^{n} p_{n,b}(\kappa)(1 - p_{n,b}(\kappa)) = a A_\gamma \kappa^{-1/\gamma}(1 - 2^{-1/\gamma})(1 + o(1)). \tag{8.4}$$

Proof. Fix a compact interval $[\kappa_-, \kappa_+] \Subset (0,\infty)$, and put

$$z_n = \frac{n}{a} \longrightarrow \infty, \qquad R_n = \big(\log(e + z_n)\big)^{4/\gamma}, \qquad B_n = \lceil a R_n \rceil.$$

Since $R_n = o(z_n)$, one has $B_n = o(n)$. Definition 4.1 therefore applies uniformly to all indices at most B_n.

Fix $0 < \varepsilon < 1 < R < \infty$. On $\varepsilon a \le b \le Ra$, the uniform convergence theorem for slowly varying functions gives

$$\frac{\lambda_{n,b}}{\lambda_{n,a}} = \left(\frac{b}{a}\right)^{\gamma} (1 + o(1))$$

uniformly. Hence the contribution of this range to the left-hand side of (8.1), divided by a, converges uniformly in $\kappa \in [\kappa_-, \kappa_+]$ to

$$\int_{\varepsilon}^{R} x^{m\gamma} e^{-\kappa x^{\gamma}}\, dx.$$

The range $b < \varepsilon a$ contains at most εa terms, and

$$\sup_{x \ge 0,\, \kappa \in [\kappa_-, \kappa_+]} x^m e^{-\kappa x} < \infty.$$

Its normalized contribution is therefore $O_m(\varepsilon)$.

It remains to control the upper tail without applying regular variation at a fixed positive fraction of n. Potter bounds, with exponent $\gamma/2$, imply that for all sufficiently large n, uniformly for $Ra < b \le B_n$,

$$c\left(\frac{b}{a}\right)^{\gamma/2} \le \frac{\lambda_{n,b}}{\lambda_{n,a}} \le C\left(\frac{b}{a}\right)^{3\gamma/2}.$$

Consequently the normalized contribution of this range is at most

$$C_m \int_{R}^{\infty} x^{3m\gamma/2} \exp(-c\kappa_- x^{\gamma/2})\, dx + o(1),$$

which tends to zero as $R \to \infty$.

For $b > B_n$, monotonicity and the preceding Potter lower bound give

$$\frac{\lambda_{n,b}}{\lambda_{n,a}} \ge \frac{\lambda_{n,B_n}}{\lambda_{n,a}} \ge cR_n^{\gamma/2} = c\big(\log(e + z_n)\big)^2.$$

Therefore

$$\frac{1}{a} \sum_{b > B_n} \left(\frac{\lambda_{n,b}}{\lambda_{n,a}}\right)^m e^{-\kappa \lambda_{n,b}/\lambda_{n,a}} \le z_n \sup_{x \ge c(\log(e+z_n))^2} x^m e^{-\kappa_- x} \longrightarrow 0.$$

Letting first $n \to \infty$, then $\varepsilon \downarrow 0$ and $R \uparrow \infty$, proves (8.1). A fixed deleted summand is bounded uniformly by $\sup_{x \ge 0} x^m e^{-\kappa_- x}$, so finitely many deletions change the sum by $O(1) = o(a)$. The special cases (8.2)–(8.3) follow from $J_{\gamma,0}(\kappa) = A_\gamma \kappa^{-1/\gamma}$ and $J_{\gamma,1}(\kappa) = A_\gamma \kappa^{-1-1/\gamma}/\gamma$. Finally, (8.4) is the difference of (8.2) at κ and at 2κ. □

Lemma 4.4 (Endpoint product envelope)**.** *Assume Definition 4.1. For every finite set $A \subset [n]$,*

$$\mathbb{P}(I_{n,a} = 1 \text{ for all } a \in A) \le C^{|A|} \prod_{a \in A} \frac{1}{a},$$

where C depends only on the constant c_Σ from Lemma 4.2.

Proof. Reveal the Luce order as a death process. When the alive set is S, the next label deleted from S is $b \in S$ with probability $\lambda_{n,b}/\sum_{c\in S}\lambda_{n,c}$. The event $I_{n,a} = 1$ says that terminal coordinate a is deleted at the transition at which the number of alive labels drops from a to $a-1$. Condition on the history before that transition. If coordinate a is already dead, the conditional probability is zero. If it is alive, the alive set has size a and contains a. Since the rates are nondecreasing, the total rate of any alive set of size a is at least $\sum_{b=1}^{a}\lambda_{n,b}$, and Lemma 4.2 gives

$$\frac{\lambda_{n,a}}{\sum_{c\in S}\lambda_{n,c}} \le \frac{1}{c_\Sigma a}.$$

Order the elements of A decreasingly and condition successively before the corresponding death levels. Multiplying the conditional bounds proves the claim. □

4.1 Separated endpoint fixed-point asymptotics

Fix $r \ge 1$ and $\eta > 0$, and let $(n, a_1, \dots, a_r) \in \mathfrak{E}_N(r,\eta)$. Put

$$B = \{a_1, \dots, a_r\}, \qquad z_j = a_j - j, \qquad 1 \le j \le r.$$

For $g = (g_1, \dots, g_r)$, set

$$y_j(g) = \log \lambda_{n,a_j} + g_j, \qquad Z_j(g) = \sum_{b\notin B} \mathbf{1}_{\{Y_{n,b} < y_j(g)\}},$$

and write

$$M_j(g_j) = \mathbb{E}Z_j(g) = \sum_{b\notin B} \exp\left(-e^{-g_j}\frac{\lambda_{n,b}}{\lambda_{n,a_j}}\right).$$

The relevant chamber is

$$\mathscr{C}_n = \{g : y_1(g) < \cdots < y_r(g)\}.$$

When $g \in \mathscr{C}_n$, put $y_0 = -\infty$, $y_{r+1} = \infty$, and define interval counts

$$W_1 = Z_1, \qquad W_j = Z_j - Z_{j-1} \quad (2 \le j \le r).$$

Thus a nonspecified priority contributes e_j to W when it lies in $[y_{j-1}, y_j)$, and contributes 0 when it lies in $[y_r, \infty)$. The target interval vector is

$$w_1 = a_1 - 1, \qquad w_j = a_j - a_{j-1} - 1 \quad (2 \le j \le r),$$

and ratio separation gives

$$c_\eta a_j \le w_j \le a_j \qquad (1 \le j \le r) \tag{8.5}$$

for all sufficiently large N.

Lemma 4.5 (Endpoint Gumbel saddle geometry)**.** *For each j, the equation*

$$M_j(g_j^*) = z_j$$

has a unique solution. Uniformly over $\mathfrak{E}_N(r,\eta)$, as $N \to \infty$,

$$g_j^* \longrightarrow g_\gamma := -\log K_\gamma, \qquad M_j'(g_j^*) \sim \frac{a_j}{\gamma}. \tag{8.6}$$

Moreover, for every fixed $L < \infty$,

$$\sup_{|g|\le L} |M_j''(g)| \le C_L a_j, \qquad c_L a_j \le M_j'(g) \le C_L a_j \tag{8.7}$$

uniformly whenever $|g - g_j^*| \le 1$. There are constants $d_\eta > 0$, $\delta_\eta > 0$, and N_0 such that, for $N \ge N_0$,

$$y_{j+1}(g_{j+1}) - y_j(g_j) \ge d_\eta \tag{8.8}$$

for every g satisfying $|g_j - g_j^*| \le \delta_\eta$, $1 \le j \le r$. In particular, this entire product neighborhood lies in $\mathscr{C}_n$.

Proof. The function M_j is continuous and strictly increasing from 0 to $n - r$, so the saddle exists uniquely. With $\kappa = e^{-g}$, Lemma 4.3, including its finite-deletion assertion, gives uniformly on compact g-sets

$$M_j(g) = a_j A_\gamma e^{g/\gamma}(1 + o(1)). \tag{8.9}$$

Differentiating term by term and applying (8.1) with $m = 1$ gives

$$\begin{aligned} M_j'(g) &= \sum_{b \notin B} e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \exp\left(-e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}}\right) \\ &= \frac{a_j A_\gamma}{\gamma} e^{g/\gamma}(1 + o(1)). \end{aligned} \tag{8.10}$$

The saddle equation and $z_j/a_j \to 1$ imply $A_\gamma e^{g_j^*/\gamma} \to 1$, which is equivalent to $g_j^* \to -\gamma \log A_\gamma = -\log K_\gamma$. Substitution in (8.10) gives the derivative asymptotic. Since

$$M_j''(g) = \sum_{b \notin B} \left[\left(e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right)^2 - e^{-g} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right] e^{-e^{-g}\lambda_{n,b}/\lambda_{n,a_j}},$$

(8.1) with $m = 1, 2$ proves the second-derivative bound. The lower and upper derivative bounds then follow from (8.10) on a compact neighborhood of the saddles.

It remains to separate the thresholds. Uniform endpoint regular variation and Potter bounds imply that there is $c_\eta > 0$ such that

$$\log \frac{\lambda_{n,a_{j+1}}}{\lambda_{n,a_j}} \ge 2c_\eta \tag{8.11}$$

for all sufficiently large N. To verify this uniformly, first restrict a_{j+1}/a_j to a fixed compact interval $[1 + \eta, R]$ and use the uniform convergence theorem; for ratios larger than R, use the Potter lower bound $\lambda_{n,a_{j+1}}/\lambda_{n,a_j} \ge c(a_{j+1}/a_j)^{\gamma/2}$, and then choose R large. Because all g_j^* converge to the same constant, the saddle threshold gaps are at least c_η for large N. Taking $\delta_\eta < c_\eta/4$ proves (8.8). □

Lemma 4.6 (Endpoint interval edge blocks)**.** *Fix $L_0 > 0$, and consider any vector $g \in \mathscr{C}_n$. Write*

$$\mathcal{I}_j(g) = [y_{j-1}(g), y_j(g)) \quad (1 \le j \le r), \qquad \mathcal{I}_0(g) = [y_r(g), \infty).$$

Let

$$m_j(g) = \mathbb{E}W_j(g).$$

For every $c_0 > 0$, there are constants $0 < \alpha < \beta < \infty$ and $c > 0$, depending only on r, η, L_0, c_0, with the following property. If $|g_j| \le L_0$ and $m_j(g) \ge c_0 a_j$, then at least ca_j indices $b \in [\alpha a_j, \beta a_j] \setminus B$ satisfy

$$\mathbb{P}(Y_{n,b} \in \mathcal{I}_j(g)) \ge c$$

and, for one fixed $\kappa(j) \in \{j+1, \dots, r, 0\}$,

$$\mathbb{P}(Y_{n,b} \in \mathcal{I}_{\kappa(j)}(g)) \ge c.$$

Here $\kappa(j) = 0$ denotes the anchor interval. The selected subblocks may be chosen disjoint as j varies, after reducing c.

In the saddle neighborhood from Lemma 4.5, one may instead choose disjoint blocks of size at least ca_j on which the two adjacent atoms e_j and e_{j+1} each have probability at least c, with $e_{r+1} = 0$. All these conclusions remain valid, with modified constants, under every interval-coordinate exponential tilt whose coordinates are bounded by a fixed constant.

Proof. The contribution to $m_j(g)$ from $b < \alpha a_j$ is at most αa_j. Since membership in the j-th interval implies $Y_{n,b} < y_j$, the contribution from $b > \beta a_j$ is bounded by

$$\sum_{b > \beta a_j} \exp\left(-e^{-g_j} \frac{\lambda_{n,b}}{\lambda_{n,a_j}}\right).$$

The moving-cutoff proof of Lemma 4.3, with a Potter bound on the intermediate tail and monotonicity beyond the moving cutoff, shows that this is $o_\beta(a_j)$, uniformly whenever $|g_j| \le L_0$. Choose α small and β large. A positive multiple of a_j of the expected interval mass then comes from $b \in [\alpha a_j, \beta a_j]$. This range contains $O(a_j)$ indices, so at least ca_j of them have j-th interval probability at least c.

On the same middle range, regular variation on fixed multiplicative intervals keeps $\lambda_{n,b}/\lambda_{n,a_j}$ between two positive constants. Since g_j is bounded, the probability $\mathbb{P}(Y_{n,b} \ge y_j)$ is also bounded below. Split this upper event among the finitely many later intervals and the anchor; the pigeonhole principle produces a common $\kappa(j) > j$, or the anchor 0, on a subblock of order a_j. Disjoint subblocks are extracted greedily. Indeed, $a_{i+1}/a_i \ge 1 + \eta$ and r is fixed, so $\sum_{i<j} a_i \le C_{r,\eta} a_j$.

In the saddle neighborhood, choose a smaller multiplicative block $[(1-\delta)a_j, (1+\delta)a_j]$, with δ small enough that these blocks are disjoint. On this block the centered threshold $y_j - \log \lambda_{n,b}$ remains in a compact interval, and (8.8) gives uniformly positive intervals on both sides of y_j. The standard Gumbel density is positive and continuous, so the immediate lower and upper interval probabilities are both bounded below. This gives the adjacent atoms. A bounded exponential tilt multiplies all probabilities in the finite categorical support by factors bounded above and below, so every two-atom lower bound persists. □

Lemma 4.7 (Endpoint Gumbel anisotropic local CLT)**.** *Let $\mathcal{U}_n = \{g : |g_j - g_j^*| \le \delta_\eta, 1 \le j \le r\}$, where δ_η is from Lemma 4.5. Let*

$$\Sigma(g) = \operatorname{Var} Z(g), \qquad D_a = \operatorname{diag}(a_1, \dots, a_r).$$

There are constants $0 < c < C < \infty$ such that, uniformly over $\mathfrak{E}_N(r, \eta)$ and $g \in \mathcal{U}_n$,

$$cD_a \le \Sigma(g) \le CD_a \tag{8.12}$$

in quadratic-form order. For every fixed $R < \infty$, uniformly for lattice points $q \in \mathbb{Z}^r$ satisfying

$$\|D_a^{-1/2}(q - M(g))\| \le R,$$

$$\begin{aligned}\mathbb{P}(Z(g) = q) &= (2\pi)^{-r/2}(\det \Sigma(g))^{-1/2} \\ &\quad \times \exp\left[-\frac{1}{2}(q - M(g))^T \Sigma(g)^{-1}(q - M(g))\right] + o\left(\prod_{j=1}^r a_j^{-1/2}\right).\end{aligned} \tag{8.13}$$

The error is uniform over the displayed family. Moreover, for every fixed $\nu < 1/6$,

$$\mathbb{P}(Z(g) = q) \le C \prod_{j=1}^r a_j^{-1/2} \exp\left[-c \sum_{j=1}^r \frac{(q_j - M_j(g_j))^2}{a_j}\right] \tag{8.14}$$

uniformly whenever $|q_j - M_j(g_j)| \le a_j^{1/2+\nu}$ for every j.

Proof. For $b \notin B$, let

$$V_b(g) = (\mathbf{1}_{\{Y_{n,b} < y_1\}}, \dots, \mathbf{1}_{\{Y_{n,b} < y_r\}}).$$

These independent vectors have the cumulative categorical support

$$(1, \dots, 1), (0, 1, \dots, 1), \dots, (0, \dots, 0, 1), 0.$$

The adjacent blocks in Lemma 4.6 imply that, for every $u \in \mathbb{R}^r$, the j-th block contributes at least $c a_j u_j^2$ to $\mathrm{Var}(u \cdot Z(g))$. Hence the covariance lower bound in (8.12) holds. The upper bound follows from $\mathrm{Var}\, Z_j(g) = O(a_j)$, which is (8.4), and Cauchy's inequality.

The same adjacent blocks give, uniformly under bounded tilts,

$$|\mathbb{E} e^{iu \cdot (Z(g) - M(g))}| \le \exp\left[-c \sum_{j=1}^r a_j \operatorname{dist}(u_j, 2\pi\mathbb{Z})^2\right]. \tag{8.15}$$

Indeed, the two adjacent cumulative atoms differ by the coordinate vector e_j, and the blocks are disjoint. Thus the lattice generated by the support differences is the full lattice $\mathbb{Z}^r$.

Fourier inversion now gives the local expansion. On the anisotropic small box $|u_j| \le R a_j^{-1/2}$, boundedness of the categorical increments and the covariance upper bound give

$$\log \mathbb{E} e^{iu \cdot (Z - M)} = -\frac{1}{2} u^T \Sigma(g) u + O\left((\max_j |u_j|) \sum_{j=1}^r a_j u_j^2\right).$$

After the change of variables $u = D_a^{-1/2} v$, the error is $O_R(a_1^{-1/2}) = o(1)$. The complement of a fixed v-ball is uniformly negligible on the product scale by (8.15). This proves (8.13).

For (8.14), apply Lemma 3.11 with $W = Z(g)$ and $A_j = a_j$. The coordinatewise Fourier hypothesis under bounded tilts follows from the adjacent blocks, and the required tilted covariance upper bound follows from bounded distortion of the categorical probabilities and the argument proving (8.12). Since $|q_j - M_j(g_j)| \le a_j^{1/2+\nu} = o(a_j)$, the truncated exponent in that lemma is exactly $(q_j - M_j(g_j))^2 / a_j$. This proves (8.14). □

Lemma 4.8 (Endpoint mixed cumulative–interval localization)**.** *There is $L_0 = L_0(r, \eta, \gamma)$ such that the following holds for $L \geq L_0$. For $g \in \mathscr{C}_n$, put*

$$\begin{aligned} B_R(g) &= \{j : g_j > L\}, \\ B_L(g) &= \{j : g_j < -L\}, \\ G(g) &= \{1, \dots, r\} \setminus (B_R(g) \cup B_L(g)), \end{aligned}$$

and let

$$H_j = z_j - M_j(g_j), \qquad H_0 = 0, \qquad d_j = H_j - H_{j-1}.$$

On every region on which the three index sets are fixed,

$$\begin{aligned} \mathbb{P}(Z(g) = z) \leq C_L \prod_{j=1}^{r} a_j^{-1/2} \exp\left[-c \sum_{j \in B_R} |H_j| - c \sum_{j \in B_L} a_j \right] \\ \times \exp\left[-c_L \sum_{j \in G} \min\left\{ \frac{H_j^2}{a_j}, a_j \right\} \right]. \end{aligned} \tag{8.16}$$

Consequently,

$$\begin{aligned} \mathbb{P}(Z(g) = z) \leq C_L \prod_{j=1}^{r} a_j^{-1/2} \exp\left[-c e^{L/(2\gamma)} \sum_{j \in B_R} a_j - c \sum_{j \in B_L} a_j \right] \\ \times \exp\left[-c_L \sum_{j \in G} a_j \min\{(g_j - g_j^*)^2, 1\} \right]. \end{aligned} \tag{8.17}$$

Here $c > 0$ is independent of L, while C_L, c_L may depend on L, r, η, γ.

Proof. Fix the offender sets and abbreviate them by B_R, B_L, G. From (8.9), uniformly for large L,

$$\begin{aligned} j \in B_R: &\quad M_j(g_j) \geq c e^{L/\gamma} a_j, \qquad H_j < 0, \\ j \in B_L: &\quad M_j(g_j) \leq C e^{-L/\gamma} a_j, \quad 0 < H_j \leq a_j. \end{aligned} \tag{8.18}$$

For $j \in G$, $g_j \in [-L, L]$, so $m_j(g) = O_L(a_j)$ and $|d_j| \leq M_L a_j$.

Let

$$K_g(\theta) = \log \mathbb{E} \exp\{\theta \cdot (W(g) - m(g))\}$$

be the centered interval-coordinate cumulant. For a cumulative-coordinate vector α, define

$$(\mathcal{T}\alpha)_k = \sum_{j \geq k} \alpha_j.$$

Because $Z_j = W_1 + \cdots + W_j$,

$$(\mathcal{T}\alpha) \cdot d = \alpha \cdot H. \tag{8.19}$$

Choose $\delta > 0$ so small that 3δ is admissible in Lemma 3.9. Set

$$\alpha_j^- = -\delta \mathbf{1}_{\{j \in B_R\}}, \qquad \alpha_j^+ = \delta \mathbf{1}_{\{j \in B_L\}}.$$

For the nonoffending interval coordinates, fix $0 < \varepsilon < c_\eta/4$, where c_η is from (8.5), and put

$$G_0 = \{j \in G : |d_j| \le \varepsilon a_j\}, \qquad G_1 = G \setminus G_0.$$

For a sufficiently small constant $\delta_L > 0$, define

$$\lambda_j = \begin{cases} \delta_L d_j/a_j, & j \in G_0, \\ \delta_L \operatorname{sgn}(d_j), & j \in G_1, \\ 0, & j \notin G. \end{cases}$$

Use the single total interval-coordinate tilt

$$\Theta = \mathcal{T}\alpha^- + \mathcal{T}\alpha^+ + \lambda.$$

Generalized Hölder inequality gives

$$K_g(\Theta) \le \frac{1}{3}K_g(3\mathcal{T}\alpha^-) + \frac{1}{3}K_g(3\mathcal{T}\alpha^+) + \frac{1}{3}K_g(3\lambda), \tag{8.20}$$

and therefore, by (8.19),

$$\begin{aligned} -\Theta \cdot d + K_g(\Theta) \le{}& \frac{1}{3}\{-3\alpha^- \cdot H + K_g(3\mathcal{T}\alpha^-)\} \\ &+ \frac{1}{3}\{-3\alpha^+ \cdot H + K_g(3\mathcal{T}\alpha^+)\} \\ &+ \frac{1}{3}\{-3\lambda \cdot d + K_g(3\lambda)\}. \end{aligned} \tag{8.21}$$

The proof of Lemma 3.9 uses only independence, nested indicator counts, and the bound that one summand contributes at most r to the cumulative linear form. It therefore applies verbatim to the present lower-priority counts. For B_R, (8.18) gives $z_j \le Ce^{-L/\gamma}M_j(g_j)$. The lower-tail part of Lemma 3.9, applied with parameter 3δ, yields

$$-3\alpha^- \cdot H + K_g(3\mathcal{T}\alpha^-) \le -c\sum_{j\in B_R} M_j(g_j) \le -c\sum_{j\in B_R} |H_j|. \tag{8.22}$$

For B_L, one has $M_j(g_j) \le Ce^{-L/\gamma}z_j$. The upper-tail part of the same lemma gives

$$-3\alpha^+ \cdot H + K_g(3\mathcal{T}\alpha^+) \le -c\sum_{j\in B_L} z_j \le -c\sum_{j\in B_L} a_j. \tag{8.23}$$

Under every bounded interval tilt ξ, the categorical support implies

$$\operatorname{Var}_\xi(v \cdot W(g)) \le C_L \sum_{j\in G} a_j v_j^2$$

for vectors supported on G. Taylor's formula for the cumulant and a sufficiently small choice of δ_L therefore give

$$-3\lambda \cdot d + K_g(3\lambda) \le -c_L \sum_{j\in G} \min\left\{\frac{d_j^2}{a_j}, a_j\right\}. \tag{8.24}$$

All coordinates of the total tilt Θ are bounded. For $j \in G_0$, $m_j(g) \geq w_j - \varepsilon a_j \geq (c_\eta/2)a_j$. Lemma 4.6 supplies forward-edge blocks of order a_j, which may be chosen disjoint. After projection onto G_0, a later atom whose coordinate is not retained becomes the anchor atom. Lemma 3.6 therefore gives under the single tilted law

$$\mathbb{P}_\Theta(W_{G_0} = w_{G_0}) \leq C_L \prod_{j \in G_0} a_j^{-1/2}. \tag{8.25}$$

The exact change-of-measure identity, together with (8.21)–(8.25), now yields the same bound as (8.16), but with interval residuals d_j in place of cumulative residuals on G, and initially only the local factors for G_0. Every index in G_1, B_R, or B_L carries a fixed exponential loss of order a_j, which absorbs its missing factor $a_j^{-1/2}$.

It remains to compare interval and cumulative residuals without discarding the boundary values. Apply Lemma 3.4 with $A_j = a_j$, with its set B_- equal to the present B_R, and with its set B_+ equal to the present B_L. Its assumptions hold by $|d_j| \leq M_L a_j$ on G and $|H_j| \leq a_j$ on B_L. Reserving a fixed fraction of the offender costs in (8.22)–(8.23) therefore gives (8.16).

Finally, on G, the derivative bounds in (8.7), monotonicity, and compactness imply

$$|H_j| \geq c_L a_j \min\{|g_j - g_j^*|, 1\}. \tag{8.26}$$

On B_R, (8.18) gives $|H_j| \geq c e^{L/\gamma} a_j$, which may be weakened to $c e^{L/(2\gamma)} a_j$. These observations convert (8.16) into (8.17). □

Lemma 4.9 (Endpoint Gumbel product localization)**.** *Uniformly over $\mathfrak{E}_N(r, \eta)$, the following assertions hold.*

(i) For every fixed compact cube $K \subset \mathbb{R}^r$ containing all saddle vectors for large N, there are $C_K, c_K > 0$ such that, throughout $K \cap \mathscr{C}_n$,

$$\mathbb{P}(Z(g) = z) \leq C_K \prod_{j=1}^{r} a_j^{-1/2} \exp\left[-c_K \sum_{j=1}^{r} a_j \min\{(g_j - g_j^*)^2, 1\}\right]. \tag{8.27}$$

(ii) With $\varphi(g) = e^{-g} e^{-e^{-g}}$,

$$\lim_{L \to \infty} \limsup_{N \to \infty} \sup_{\mathfrak{E}_N(r,\eta)} \frac{\displaystyle\int_{\mathscr{C}_n \cap \{g \notin [-L,L]^r\}} \prod_{j=1}^{r} \varphi(g_j) \mathbb{P}(Z(g) = z)\, dg}{\prod_{j=1}^{r} a_j^{-1}} = 0. \tag{8.28}$$

Proof. Choose L so that $K \subset [-L, L]^r$. With no offenders, the proof of Lemma 4.8 gives (8.27); equivalently, apply (8.16) with $G = \{1, \dots, r\}$.

For the tail, partition the integration region according to the finitely many sets B_R, B_L, G, and use (8.17). Every nonoffending coordinate contributes a factor $O(a_j^{-1/2})$ after integration on its Gaussian scale, in addition to the prefactor $a_j^{-1/2}$. For a left offender, $e^{-c a_j}$ absorbs the missing local factor and

$$\int_{-\infty}^{-L} \varphi(g)\, dg = e^{-e^L}.$$

For a right offender, $e^{-ce^{L/(2\gamma)}a_j}$ absorbs the missing local factor and

$$\int_L^\infty \varphi(g)\,dg = 1 - e^{-e^{-L}} \le e^{-L}.$$

At least one offender is present. Summing over the finitely many offender sets proves (8.28). □

Lemma 4.10 (Endpoint covariance stability on the saddle scale)**.** *Let*

$$H_a = \operatorname{diag}(a_1^{-1/2}, \dots, a_r^{-1/2}), \qquad \Sigma_* = \Sigma(g^*).$$

For every fixed $R < \infty$,

$$\sup_{\|u\| \le R} \left\| D_a^{-1/2} \left[\Sigma(g^* + H_a u) - \Sigma_* \right] D_a^{-1/2} \right\|_{\mathrm{op}} \longrightarrow 0 \tag{8.29}$$

uniformly over $\mathfrak{E}_N(r, \eta)$. Consequently, uniformly on the same sets,

$$\frac{\det \Sigma(g^* + H_a u)}{\det \Sigma_*} \longrightarrow 1 \tag{8.30}$$

and

$$\left\| D_a^{1/2} \left[\Sigma(g^* + H_a u)^{-1} - \Sigma_*^{-1} \right] D_a^{1/2} \right\|_{\mathrm{op}} \longrightarrow 0. \tag{8.31}$$

Proof. For $j \le k$, put

$$p_{b,j}(g_j) = \exp\left(-e^{-g_j} \frac{\lambda_{n,b}}{\lambda_{n,a_j}} \right).$$

Since the threshold events are nested in the chamber,

$$\Sigma_{jk}(g) = \sum_{b \notin B} p_{b,j}(g_j)(1 - p_{b,k}(g_k)). \tag{8.32}$$

Lemma 4.3, with $m = 1$, gives uniformly on a fixed saddle neighborhood

$$|\partial_{g_j} \Sigma_{jk}(g)| \le C a_j, \qquad |\partial_{g_k} \Sigma_{jk}(g)| \le C a_k. \tag{8.33}$$

The same estimate, with the evident interpretation, holds on the diagonal. Hence, for $\|u\| \le R$,

$$|\Sigma_{jk}(g^* + H_a u) - \Sigma_{jk}(g^*)| \le C_R(\sqrt{a_j} + \sqrt{a_k}).$$

After division by $\sqrt{a_j a_k}$, this is at most $C_R(a_j^{-1/2} + a_k^{-1/2}) \le 2C_R a_1^{-1/2}$. Since r is fixed, (8.29) follows. By (8.12), the normalized matrices $D_a^{-1/2}\Sigma(g)D_a^{-1/2}$ remain in a compact subset of the positive definite matrices. Continuity of determinant and inversion on this set gives (8.30) and (8.31). □

Proposition 4.11 (Separated endpoint local estimate)**.** *Assume Definition 4.1. Fix $r \ge 1$ and $\eta > 0$. Then*

$$\lim_{N \to \infty} \sup_{(n,a) \in \mathfrak{E}_N(r,\eta)} \left| \frac{\mathbb{P}(I_{n,a_1} = \cdots = I_{n,a_r} = 1)}{c_\gamma^r / (a_1 \cdots a_r)} - 1 \right| = 0. \tag{8.34}$$

Proof. On the joint fixed-point event, the specified priorities are necessarily ordered as $y_1 < \cdots < y_r$. Conditional on their centered Gumbels g in this chamber, exactly $j-1$ specified priorities lie below y_j, so the event is equivalent to $Z_j(g) = a_j - j$, $1 \le j \le r$. Therefore

$$\begin{aligned}&\mathbb{P}(I_{n,a_1} = \cdots = I_{n,a_r} = 1)\\ &\quad = \int_{\mathscr{C}_n} \prod_{j=1}^{r} \varphi(g_j)\, \mathbb{P}(Z(g) = z)\, dg.\end{aligned} \tag{8.35}$$

Lemma 4.5 puts a fixed product neighborhood of g^* inside the chamber. Lemma 4.9 removes the noncompact region on the scale $\prod_j a_j^{-1}$, and its compact bound makes the part outside any fixed saddle neighborhood exponentially negligible on that scale.

Set

$$D_n = \operatorname{diag}(M_1'(g_1^*), \dots, M_r'(g_r^*)), \qquad g = g^* + H_a u.$$

For bounded u, Taylor's theorem and (8.7) give

$$M(g) - z = D_n H_a u + r_n(u), \qquad \|D_a^{-1/2} r_n(u)\| \longrightarrow 0 \tag{8.36}$$

uniformly on fixed u-balls. Indeed, the j-th remainder is $O_R(1)$, hence is $o(\sqrt{a_j})$. The specified density satisfies

$$\prod_{j=1}^{r} \varphi(g_j^* + u_j/\sqrt{a_j}) = \prod_{j=1}^{r} \varphi(g_j^*)(1 + o(1)) \tag{8.37}$$

uniformly on fixed balls.

Apply Lemma 4.7, then use Lemma 4.10 to replace $\Sigma(g)$ by Σ_*. On every fixed u-ball, the integrand in (8.35), including the Jacobian $|\det H_a|$, is

$$\begin{aligned}&(1 + o(1)) \prod_{j=1}^{r} \varphi(g_j^*)(2\pi)^{-r/2} (\det \Sigma_*)^{-1/2} |\det H_a|\\ &\quad \times \exp\left[-\frac{1}{2} u^T H_a D_n \Sigma_*^{-1} D_n H_a u\right] du.\end{aligned} \tag{8.38}$$

The matrices in the exponent are uniformly positive definite and uniformly bounded. The compact localization estimate (8.27), after the change of variables, provides the integrable majorant

$$C \left(\prod_{j=1}^{r} a_j^{-1}\right) \exp\left[-c \sum_{j=1}^{r} \min\{u_j^2, a_j\}\right].$$

Thus the fixed-ball limit may be extended uniformly to $\mathbb{R}^r$. The Gaussian determinant cancels exactly:

$$\begin{aligned}&(2\pi)^{-r/2} (\det \Sigma_*)^{-1/2} |\det H_a| \int_{\mathbb{R}^r} e^{-\frac{1}{2} u^T H_a D_n \Sigma_*^{-1} D_n H_a u}\, du\\ &\quad = |\det D_n|^{-1}.\end{aligned} \tag{8.39}$$

Consequently,

$$\mathbb{P}(I_{n,a_1} = \cdots = I_{n,a_r} = 1) = (1 + o(1)) \prod_{j=1}^{r} \frac{\varphi(g_j^*)}{M_j'(g_j^*)}. \tag{8.40}$$

By (8.6),

$$\varphi(g_j^*) \longrightarrow K_\gamma e^{-K_\gamma}, \qquad M_j'(g_j^*) \sim a_j/\gamma.$$

Each factor in (8.40) is therefore asymptotic to c_γ/a_j, uniformly over the admissible family. This proves (8.34). □

Theorem 4.12 (Endpoint fixed-point universality)**.** *Assume the terminal rate array is uniformly endpoint regularly varying with index $\gamma > 0$ in the sense of Definition 4.1. Let*

$$\Xi_n = \sum_{a=1}^{n} I_{n,a}\delta_{\log(a/b_n)},$$

where $b_n \to \infty$ and $b_n = o(n)$. For every bounded interval $J \subset \mathbb{R}$, the restriction $\Xi_n|_J$, regarded as a random locally finite counting measure with the vague topology, satisfies

$$\Xi_n|_J \Rightarrow \mathrm{PPP}(c_\gamma\, dx)|_J. \tag{8.41}$$

Equivalently, counts in finitely many disjoint bounded intervals converge jointly to independent Poisson random variables with means c_γ times their lengths. Moreover, for every fixed $r \geq 1$,

$$\mathbb{E}(F_n)_r \sim (c_\gamma \log n)^r. \tag{8.42}$$

In particular,

$$\mathbb{E}C_1(\sigma_n) \sim c_\gamma \log n, \qquad \frac{C_1(\sigma_n)}{c_\gamma \log n} \longrightarrow 1 \quad \textit{in probability.} \tag{8.43}$$

Proof. We first prove the local point-process assertion. For every bounded interval $I = (u, v]$, the elementary logarithmic Riemann sum gives

$$\sum_{\log(a/b_n)\in I} \frac{1}{a} \longrightarrow v - u = |I|. \tag{8.44}$$

All coordinates in this sum tend to infinity and are $o(n)$. Fix disjoint bounded intervals $I_1, \ldots, I_m$ and nonnegative integers $k_1, \ldots, k_m$, and put $k = \sum_\ell k_\ell$. Expand

$$\mathbb{E}\prod_{\ell=1}^{m}(\Xi_n(I_\ell))_{k_\ell}$$

over ordered distinct terminal coordinates, colored according to their intervals. Restrict first to tuples whose logarithms are pairwise more than η apart. After sorting the coordinates, Proposition 4.11 applies with ratio-separation parameter $e^\eta - 1$, uniformly over the entire restricted sum. Its contribution is therefore

$$c_\gamma^k(1 + o(1)) \sum_{\eta\text{-separated}}^{*} \frac{1}{a_1 \cdots a_k}. \tag{8.45}$$

The logarithmic diagonal neighborhood has negligible harmonic weight as $\eta \downarrow 0$. Indeed, uniformly for a in a fixed multiplicative window,

$$\sum_{b:\,|\log b - \log a| \leq \eta} \frac{1}{b} \leq C\eta + o(1), \tag{8.46}$$

and hence a union bound over pairs gives an $O_{k,I_1,\dots,I_m}(\eta)+o(1)$ bound for the weighted sum of excluded tuples. Lemma 4.4 gives the same bound, up to a fixed constant, for their probability contribution. Using (8.44) in (8.45) and then letting $\eta\downarrow 0$, we obtain

$$\mathbb{E}\prod_{\ell=1}^{m}(\Xi_n(I_\ell))_{k_\ell}\longrightarrow\prod_{\ell=1}^{m}(c_\gamma|I_\ell|)^{k_\ell}. \tag{8.47}$$

These are the mixed factorial moments of independent Poisson variables, so the interval-count vectors converge to the asserted independent Poisson vector.

We now upgrade count convergence to convergence of point measures. By the one-point case of Lemma 4.4,

$$\sup_n \mathbb{E}\Xi_n(J)\le C\sup_n\sum_{\log(a/b_n)\in J}\frac{1}{a}<\infty \tag{8.48}$$

for every bounded J, after ignoring finitely many initial n. Let $f\ge 0$ be continuous with compact support in J. Choose a finite partition of J into half-open intervals and a step function f_π, constant on the partition cells, with $\|f-f_\pi\|_\infty\to 0$ as the mesh tends to zero. Since $|e^{-x}-e^{-y}|\le|x-y|$ for $x,y\ge 0$,

$$\left|\mathbb{E}e^{-\langle\Xi_n,f\rangle}-\mathbb{E}e^{-\langle\Xi_n,f_\pi\rangle}\right|\le\|f-f_\pi\|_\infty\mathbb{E}\Xi_n(J), \tag{8.49}$$

uniformly in n. For fixed π, the interval-count convergence just proved gives

$$\mathbb{E}e^{-\langle\Xi_n,f_\pi\rangle}\longrightarrow\exp\left[-c_\gamma\int_J(1-e^{-f_\pi(x)})\,dx\right].$$

Letting the mesh tend to zero in view of (8.49) yields

$$\mathbb{E}e^{-\langle\Xi_n,f\rangle}\longrightarrow\exp\left[-c_\gamma\int_J(1-e^{-f(x)})\,dx\right], \tag{8.50}$$

the Laplace functional of $\mathrm{PPP}(c_\gamma\,dx)|_J$. Laplace functionals on nonnegative compactly supported continuous functions determine the law of a locally finite counting measure and are convergence determining for the vague topology. This proves (8.41); the uniform first-moment bound (8.48) also gives tightness directly.

It remains to prove the full factorial moments. Put

$$L_n=\lceil\log n\rceil,\qquad U_n=\lfloor n/\log n\rfloor,\qquad H_n=\sum_{L_n\le a\le U_n}\frac{1}{a}.$$

Then $H_n\sim\log n$. Fix once and for all a positive separation η_0. The harmonic weight of ordered r-tuples in $[L_n,U_n]$ containing a pair with $|\log a_i-\log a_j|\le\eta_0$ is $O_r(H_n^{r-1})$. To see this, fix one coordinate a_i, use

$$\sum_{b:\,|\log b-\log a_i|\le\eta_0}\frac{1}{b}\le C_{\eta_0},$$

and sum the remaining $r-2$ coordinates freely. Repeated coordinates are included in this bound. Therefore the separated weighted sum is

$$H_n^r+O_r(H_n^{r-1}). \tag{8.51}$$

On the separated tuples, Proposition 4.11 applies uniformly and gives the contribution

$$c_\gamma^r H_n^r + o(H_n^r).$$

The nonseparated probability contribution is $O_r(H_n^{r-1})$ by Lemma 4.4 and the preceding harmonic estimate. Thus, for $G_n = \sum_{L_n \le a \le U_n} I_{n,a}$,

$$\mathbb{E}(G_n)_r \sim (c_\gamma \log n)^r. \tag{8.52}$$

Finally, the harmonic mass outside $[L_n, U_n]$ is

$$H_n^{\mathrm{out}} := \sum_{a<L_n} \frac{1}{a} + \sum_{a>U_n} \frac{1}{a} = O(\log\log n).$$

By the product envelope, the total contribution to $\mathbb{E}(F_n)_r$ from ordered tuples containing at least one omitted coordinate is at most

$$C_r H_n^{\mathrm{out}} \left(\sum_{a=1}^{n} \frac{1}{a}\right)^{r-1} = O_r((\log n)^{r-1} \log\log n) = o((\log n)^r).$$

Combining this with (8.52) proves (8.42).

Taking $r = 1, 2$ gives $\mathbb{E}F_n \sim c_\gamma \log n$ and $\mathrm{Var}(F_n) = o((\log n)^2)$. Chebyshev's inequality yields (8.43). □

4.2 Further endpoint limits

The same regular-variation calculation also gives two boundary limits that are useful for interpreting the endpoint process. The first is a mesoscopic one-coordinate displacement law; the second is a microscopic infinite random permutation.

Theorem 4.13 (Endpoint displacement law)**.** *Assume Definition 4.1. Let $a_{n,1}, \dots, a_{n,m}$ be distinct terminal coordinates, where m is fixed, $a_{n,j} \to \infty$, and $a_{n,j} = o(n)$ for every j. Then*

$$\left(\frac{\tau_n(a_{n,1})}{a_{n,1}}, \dots, \frac{\tau_n(a_{n,m})}{a_{n,m}}\right) \Rightarrow \left(A_\gamma E_1^{-1/\gamma}, \dots, A_\gamma E_m^{-1/\gamma}\right),$$

where $E_1, \dots, E_m$ are independent Exp(1) *variables and $A_\gamma = \Gamma(1 + 1/\gamma)$. Equivalently,*

$$\log \frac{\tau_n(a_{n,j})}{a_{n,j}} \Rightarrow X_\gamma,$$

where X_γ has density

$$k_\gamma(x) = \gamma K_\gamma e^{-\gamma x} \exp\{-K_\gamma e^{-\gamma x}\}, \qquad K_\gamma = A_\gamma^\gamma.$$

Proof. Write

$$Y_{n,a} = \log \lambda_{n,a} + G_a,$$

with independent standard Gumbels. Conditional on $G_a = g$,

$$\tau_n(a) - 1 = \sum_{b \ne a} \mathbf{1}_{\{Y_{n,b} < Y_{n,a}\}}$$

is a sum of independent Bernoulli variables with mean

$$M_{n,a}(g) = \sum_{b \neq a} \exp\left(-e^{-g}\frac{\lambda_{n,b}}{\lambda_{n,a}}\right).$$

Lemma 4.3 gives, uniformly for g in compact sets,

$$\frac{M_{n,a}(g)}{a} \longrightarrow A_\gamma e^{g/\gamma}.$$

The conditional variance is at most the conditional mean, hence is $O(a)$ on compact g-sets. Chebyshev's inequality therefore gives, uniformly on such sets,

$$\frac{\tau_n(a)}{a} - A_\gamma e^{G_a/\gamma} \longrightarrow 0 \quad \text{in probability.}$$

Truncating the Gumbel variables to a fixed compact set and then letting the truncation grow proves the one-coordinate assertion. For finitely many specified coordinates, delete them from the Bernoulli sums. Lemma 4.3 is unchanged after finitely many deletions, and a union bound over the conditional Chebyshev estimates proves the joint convergence. The limiting variables are independent because the specified Gumbels are independent.

Finally, if $E = e^{-G} \sim \mathrm{Exp}(1)$, then $A_\gamma e^{G/\gamma} = A_\gamma E^{-1/\gamma}$. The change of variables $x = \log A_\gamma + G/\gamma$ gives the displayed density k_γ. □

4.3 Mesoscopic endpoint tangent measure

The microscopic limit in Theorem 4.19 is an infinite random permutation. On every mesoscopic scale containing $b_n \to \infty$ terminal coordinates, the correct object is instead a dense normalized graph measure. The unnormalized graph cannot have a locally finite point-process limit, because a compact source window contains order b_n points.

Define

$$\mathcal{K}_\gamma(x,y) = \frac{1}{y}k_\gamma\Big(\log\frac{y}{x}\Big) = \gamma K_\gamma \frac{x^\gamma}{y^{\gamma+1}} \exp\left\{-K_\gamma\left(\frac{x}{y}\right)^\gamma\right\}, \qquad x,y>0.$$

Theorem 4.14 (Mesoscopic endpoint tangent measure)**.** *Assume Definition 4.1. Let $b_n \to \infty$ with $b_n = o(n)$, and put*

$$\mathcal{M}_n^{(b_n)} = \frac{1}{b_n}\sum_{a=1}^{n} \delta_{(a/b_n,\,\tau_n(a)/b_n)}.$$

Each $\mathcal{M}_n^{(b_n)}$ is a Radon measure on compact subsets of $(0,\infty)^2$, and the density $\mathcal{K}_\gamma$ is locally integrable there. Then, vaguely in probability on $(0,\infty)^2$,

$$\mathcal{M}_n^{(b_n)} \longrightarrow \mathcal{K}_\gamma(x,y)\,dx\,dy.$$

More precisely, for every $\varphi \in C_c((0,\infty)^2)$,

$$\mathbb{E}\left|\langle \mathcal{M}_n^{(b_n)}, \varphi\rangle - \int_0^\infty\int_0^\infty \varphi(x,y)\mathcal{K}_\gamma(x,y)\,dy\,dx\right| \longrightarrow 0.$$

The limiting kernel is bistochastic with respect to Lebesgue measure.

Lemma 4.15 (Uniform mesoscopic Laplace sums)**.** *Fix $0 < x_- < x_+ < \infty$ and $0 < u_- < u_+ < \infty$. Uniformly for $x_- b_n \le a \le x_+ b_n$ and $u \in [u_-, u_+]$,*

$$\frac{1}{b_n} \sum_{c=1}^{n} \exp\left\{-u \frac{\lambda_{n,c}}{\lambda_{n,a}}\right\} = \frac{a}{b_n} A_\gamma u^{-1/\gamma} + o(1).$$

For every fixed $u_- > 0$, the left-hand side is uniformly bounded for all $u \ge u_-$.

Proof. Put $X_n = n/b_n \to \infty$ and choose

$$R_n = (\log(e + X_n))^{4/\gamma}, \qquad B_n = \lceil 2x_+ b_n R_n \rceil.$$

Then $B_n = o(n)$. Uniform endpoint regular variation applies to all $a \in [x_- b_n, x_+ b_n]$ and $c \le B_n$. On every fixed multiplicative interval $\delta a \le c \le Ra$, the uniform convergence theorem for slowly varying functions gives

$$\lambda_{n,c}/\lambda_{n,a} = (c/a)^\gamma (1 + o(1))$$

uniformly. Hence

$$\frac{1}{b_n} \sum_{\delta a \le c \le Ra} e^{-u\lambda_{n,c}/\lambda_{n,a}} = \frac{a}{b_n} \int_\delta^R e^{-uv^\gamma}\, dv + o(1).$$

The range $c < \delta a$ contributes at most $x_+ \delta$. Potter's bound with exponent $\gamma/2$ gives, for $Ra < c \le B_n$,

$$\lambda_{n,c}/\lambda_{n,a} \ge c(c/a)^{\gamma/2},$$

so the normalized contribution is bounded by

$$C \frac{a}{b_n} \int_R^\infty e^{-cu_- v^{\gamma/2}}\, dv.$$

For $c > B_n$, monotonicity and Potter's bound give

$$\lambda_{n,c}/\lambda_{n,a} \ge c R_n^{\gamma/2},$$

and therefore

$$\frac{1}{b_n} \sum_{c > B_n} e^{-u\lambda_{n,c}/\lambda_{n,a}} \le X_n \exp\{-cu_- R_n^{\gamma/2}\} = o(1).$$

Let first $n \to \infty$, then $\delta \downarrow 0$ and $R \uparrow \infty$. Since

$$\int_0^\infty e^{-uv^\gamma}\, dv = A_\gamma u^{-1/\gamma},$$

the first assertion follows. Monotonicity in u gives the final bound. □

Let $E_{n,a} = \lambda_{n,a} T_{n,a}$. These variables are independent and standard exponential. Conditional on $E_{n,a} = u$,

$$\tau_n(a) = 1 + \sum_{c \ne a} \xi_{n,a,c}(u), \qquad \mathbb{P}(\xi_{n,a,c}(u) = 1) = e^{-u\lambda_{n,c}/\lambda_{n,a}},$$

with independent summands. Put

$$r_{n,a}(u) = 1 + \sum_{c \ne a} e^{-u\lambda_{n,c}/\lambda_{n,a}}.$$

Lemma 4.16 (Conditional rank self-averaging)**.** *Fix $0 < x_- < x_+ < \infty$. For every $\varepsilon, \delta > 0$,*

$$\sup_{x_- b_n \le a \le x_+ b_n} \mathbb{P}\big(|\tau_n(a) - r_{n,a}(E_{n,a})| > \delta b_n\big) \le \varepsilon + \frac{C_\varepsilon}{\delta^2 b_n}$$

for all large n. Moreover, uniformly for $a/b_n \in [x_-, x_+]$ and u in a compact subset of $(0,\infty)$,

$$\frac{r_{n,a}(u)}{b_n} \longrightarrow \frac{a}{b_n} A_\gamma u^{-1/\gamma}.$$

Proof. Conditionally on $E_{n,a} = u$,

$$\mathrm{Var}(\tau_n(a) \mid E_{n,a} = u) \le r_{n,a}(u).$$

On $\{E_{n,a} \ge \varepsilon\}$, Lemma 4.15 and monotonicity in u give $r_{n,a}(E_{n,a}) \le C_\varepsilon b_n$. Conditional Chebyshev therefore bounds the deviation probability on this event by $C_\varepsilon/(\delta^2 b_n)$. The complementary event has probability at most ε. The second assertion follows immediately from Lemma 4.15; omitting the $c = a$ summand and adding the leading one changes the sum by $O(1)$. □

Proof of Theorem 4.14. Fix $\varphi \in C_c((0,\infty)^2)$, and choose $0 < x_- < x_+ < \infty$ containing the first-coordinate projection of its support. Set

$$Z_n(\varphi) = \frac{1}{b_n} \sum_a \varphi(a/b_n, \tau_n(a)/b_n)$$

and

$$\widehat{Z}_n(\varphi) = \frac{1}{b_n} \sum_a \varphi(a/b_n, r_{n,a}(E_{n,a})/b_n).$$

Only $O(b_n)$ summands are nonzero. Uniform continuity of φ and Lemma 4.16 give

$$\mathbb{E}|Z_n(\varphi) - \widehat{Z}_n(\varphi)| \longrightarrow 0.$$

Indeed, split according as the two second coordinates differ by at most δ, use the modulus of continuity on that event, and then let $n \to \infty$, $\delta \downarrow 0$, and $\varepsilon \downarrow 0$.

The summands in $\widehat{Z}_n(\varphi)$ are independent functions of the independent variables $E_{n,a}$. Hence

$$\mathrm{Var}(\widehat{Z}_n(\varphi)) \le C_\varphi / b_n \longrightarrow 0.$$

It remains to compute the expectations. If $U \sim \mathrm{Exp}(1)$, Lemma 4.16, uniformly on compact u-sets, gives

$$\mathbb{E}\widehat{Z}_n(\varphi) \longrightarrow \int_0^\infty \int_0^\infty \varphi(x, x A_\gamma u^{-1/\gamma}) e^{-u}\, du\, dx.$$

To justify the passage, truncate u to $[\varepsilon, M]$, use uniform convergence and an ordinary Riemann sum, and then use the exponential mass of $(0,\varepsilon) \cup (M,\infty)$.

For fixed x, make the change of variables $y = x A_\gamma u^{-1/\gamma}$. Then $u = K_\gamma (x/y)^\gamma$ and

$$e^{-u} \left| \frac{du}{dy} \right| = \mathcal{K}_\gamma(x, y).$$

This proves the L^1 convergence of test-function integrals. A countable convergence-determining subset of $C_c((0,\infty)^2)$ then gives vague convergence in probability.

Finally, the first marginal identity follows from the same change of variables: $\mathcal{K}_\gamma(x,\cdot)$ is the density of $xA_\gamma U^{-1/\gamma}$. For the second marginal, put $v = K_\gamma(x/y)^\gamma$ to obtain

$$\int_0^\infty \mathcal{K}_\gamma(x,y)\,dx = K_\gamma^{-1/\gamma}\int_0^\infty v^{1/\gamma}e^{-v}\,dv = 1.$$

Thus the kernel is bistochastic. □

Corollary 4.17 (Rectangle law)**.** *For bounded intervals $I, J \Subset (0,\infty)$,*

$$\frac{1}{b_n}\#\left\{a : a/b_n \in I,\ \tau_n(a)/b_n \in J\right\} \xrightarrow{\mathbb{P}} \int_I\int_J \mathcal{K}_\gamma(x,y)\,dy\,dx.$$

Proof. The boundary of $I \times J$ has zero mass under the continuous limiting density, so the assertion follows from vague convergence and approximation by compactly supported continuous functions. □

Proposition 4.18 (Multiplicative semigroup and diagonal traces)**.** *For every $r \geq 1$,*

$$\mathcal{K}_\gamma^r(x,y) = \frac{1}{y}k_\gamma^{*r}\left(\log\frac{y}{x}\right).$$

In particular,

$$\mathcal{K}_\gamma^r(x,x) = \frac{1}{x}k_\gamma^{*r}(0).$$

Proof. For $r = 2$, substitute $z = xe^w$ in the composition integral:

$$\int_0^\infty \mathcal{K}_\gamma(x,z)\mathcal{K}_\gamma(z,y)\,dz = \frac{1}{y}\int_{\mathbb{R}} k_\gamma(w)k_\gamma\left(\log\frac{y}{x} - w\right)\,dw.$$

This is $y^{-1}k_\gamma^{*2}(\log(y/x))$. Iterate. □

4.4 Microscopic endpoint permutation

Theorem 4.19 (Microscopic endpoint permutation)**.** *Assume Definition 4.1. Extend the finite permutation $\tau_n \in S_n$ to a permutation $\bar\tau_n$ of $\mathbb{N}$ by*

$$\bar\tau_n(a) = \begin{cases}\tau_n(a), & a \leq n,\\ a, & a > n.\end{cases}$$

Then, in the product topology on $\mathbb{N}^{\mathbb{N}}$,

$$\bar\tau_n \Rightarrow \tau^{(\gamma)}, \tag{8.53}$$

where, for independent $E_1, E_2, \ldots \sim \mathrm{Exp}(1)$,

$$\tau^{(\gamma)}(a) = 1 + \#\left\{b \neq a : \frac{E_b}{b^\gamma} > \frac{E_a}{a^\gamma}\right\}.$$

Almost surely, $\tau^{(\gamma)}$ is a permutation of $\mathbb{N}$.

Proof. Normalize by the first terminal rate and put

$$r_{n,b} = \frac{\lambda_{n,b}}{\lambda_{n,1}}, \qquad Q_{n,b} = \lambda_{n,1} T_{n,b} = \frac{E_b}{r_{n,b}},$$

using independent $E_b \sim \mathrm{Exp}(1)$. For every fixed B, uniform endpoint regular variation and the uniform convergence theorem for slowly varying functions give

$$(r_{n,1}, \dots, r_{n,B}) \longrightarrow (1, 2^\gamma, \dots, B^\gamma).$$

Hence, under this coupling,

$$(Q_{n,1}, \dots, Q_{n,B}) \longrightarrow (E_1, E_2/2^\gamma, \dots, E_B/B^\gamma)$$

almost surely.

We remove the tail uniformly. Fix m and $\varepsilon > 0$. For fixed $a \le m$, the ratios $r_{n,a}$ remain bounded, so

$$\mathbb{P}(Q_{n,a} < \varepsilon) \le C_m \varepsilon.$$

Choose $W_n = \lfloor n/\log n \rfloor$. Potter's bound, applied with reference coordinate 1, gives

$$r_{n,b} \ge c b^{\gamma/2}, \qquad B < b \le W_n,$$

for all sufficiently large n. By monotonicity, the same lower bound at $b = W_n$ applies to every $b > W_n$. Consequently,

$$\begin{aligned} \sum_{b>B} \mathbb{P}(Q_{n,b} > \varepsilon) &= \sum_{b>B} e^{-\varepsilon r_{n,b}} \\ &\le \sum_{b>B} e^{-c\varepsilon b^{\gamma/2}} + n e^{-c\varepsilon W_n^{\gamma/2}}. \end{aligned} \tag{8.54}$$

The second term tends to zero, and the first tends to zero as $B \to \infty$, uniformly for all sufficiently large n. Thus, outside an event of probability at most $C_m\varepsilon + o_B(1)$, every comparison determining the ranks of the first m coordinates is decided by the first B clocks. The same argument applies to the limiting clocks E_b/b^γ. Finite collections of all these clocks have no ties. Taking first $n \to \infty$, then $B \to \infty$, and finally $\varepsilon \downarrow 0$, proves finite-dimensional convergence of $(\bar\tau_n(1), \dots, \bar\tau_n(m))$ for every fixed m, which is (8.53).

For every $t > 0$,

$$\sum_{b=1}^{\infty} \mathbb{P}(E_b/b^\gamma > t) = \sum_{b=1}^{\infty} e^{-b^\gamma t} < \infty.$$

Borel–Cantelli implies that only finitely many limiting clocks exceed any positive threshold. Moreover,

$$\mathbb{P}\left(\sup_{b \ge 1} \frac{E_b}{b^\gamma} > t \right) \le \sum_{b \ge 1} e^{-b^\gamma t} \longrightarrow 0 \qquad (t \to \infty),$$

so the clock set has a finite maximum. The clocks are almost surely distinct, and zero is their only accumulation point. Their decreasing enumeration therefore contains every label exactly once and assigns every positive integer rank exactly once. Thus $\tau^{(\gamma)}$ is almost surely a permutation of $\mathbb{N}$. □

Remark 4.20 (The Gumbel origin of the constant). The preceding proof makes the constant transparent. Conditional on $Y_{n,a} = \log \lambda_{n,a} + g$, the expected number of lower terminal priorities is

$$\sum_b \exp\left(-e^{-g}\frac{\lambda_{n,b}}{\lambda_{n,a}}\right) \sim aA_\gamma e^{g/\gamma}.$$

The fixed-point saddle is therefore $A_\gamma e^{g/\gamma} = 1$, so $e^{-g} = K_\gamma = A_\gamma^\gamma$. The density of the specified centered Gumbel at the saddle is $K_\gamma e^{-K_\gamma}$, while the derivative of the mean with respect to g is a/γ. Their quotient is exactly c_γ/a, with

$$c_\gamma = \gamma K_\gamma e^{-K_\gamma}.$$

For $\gamma = 1$, $A_\gamma = K_\gamma = 1$, and the constant reduces to the Sukhatme value e^{-1}.

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